



Flying Liferaft

Propeller-Engine Performance, Systems Layout and Inertial Estimations



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Abstract

There had been over 30000 migrant disappearances on average each year in the Mediterranean sea [1]. Therefore, a flying life raft solution was proposed. This report goes through the propulsion system requirements, design and performance of the flying life raft design. It was found that for a design with a 51.5kg, the minimum power requirements would be 6.54 kW. The designed pusher and VTOL prop-engine was found to have a spool up time of 10seconds and 20seconds respectively. This report also outlines the layout of the flight control system. Avionics, including the microcontrollers and communications system for the aircraft were selected to optimise the aircraft performance. Lastly, detailed weight and balancing estimations suggest that the aircraft would have a MTOW of around 49.5kg instead. The initial estimates were largely unreliable due to large component estimate differences. The aircraft meets the design requirements for the most part. In order to achieve the same mission range in the conceptual design stage, extra fuel would need to be carried due to the increase in the estimated drag of the aircraft. A few other issues were identified and should be fixed in the future iterations.

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List of Symbols

a	axial induction factor
a'	tangential induction factor
$b_{friction}$	motor frictional constant
c	sectional chord
C_D	drag coefficient
C_d	sectional drag coefficient
C_L	lift coefficient
C_l	sectional lift coefficient
f_{drag}	propeller drag-induced-torque
f_{mounts}	mounting weight ratio
f_{wiring}	wiring weight ratio
H	Antenna height from the ground
h	flight altitude
I_{x0}	moment of inertia of the aircraft about the x-axis
I_{y0}	moment of inertia of the aircraft about the y-axis
I_{z0}	moment of inertia of the aircraft about the z-axis
I_{xz0}	moment of inertia of the aircraft about the x-axis
$I_x(i)$	moment of inertia of the components about the x-axis
$I_y(i)$	moment of inertia of the components about the y-axis
$I_z(i)$	moment of inertia of the aircraft about the z-axis
$J_{propeller}$	moment of inertia of the propeller
r	distance from the propeller element from the centre
r_f	propeller tip radius
r_0	propeller root radius
Re	Reynolds number
T	Torque (engine/motor)
W_{Tot}	total weight of the aircraft
$W_{avionics}$	avionics weight
W_{comms}	communication systems weight
W_{comp}	aircraft mountable components weight
$W_{comp,mount}$	mounting weight of the component
$W_{payload}$	payload weight
W_{wiring}	aircraft cabling weight
W_{Tot}	total weight of the aircraft
$W(i)$	weight of the component
X_{CG}	x position of the centre of gravity
Y_{CG}	y position of the centre of gravity
Z_{CG}	z position of the centre of gravity
$x(i)$	x position of the component
$y(i)$	y position of the component
$z(i)$	z position of the component
α	angle of attack
β	twist angle
ρ	air density at mean sea level
$\ddot{\theta}$	angular acceleration
$\dot{\theta}$	angular velocity
μ	dynamic viscosity

1 Introduction

1.1 Task overview

The flying life raft was aimed to deliver rapid, self-inflating floatation devices with a time on target of 30 minutes to reduce migrant deaths. To achieve such design specifications, this report evaluated the propulsion system requirement, performance, control system and layout design of the flying life raft solution.

2 Initial sizing

From the constraints diagram in [2], the aircraft has a wing loading and thrust to weight ratio of 1006 N/m^2 and 0.22 respectively. The MTOW of the aircraft was estimated to be 51.5kg. In order to size the battery, the the power requirement of the aircraft for VTOL operation would need to be evaluated.

2.1 VTOL power estimation

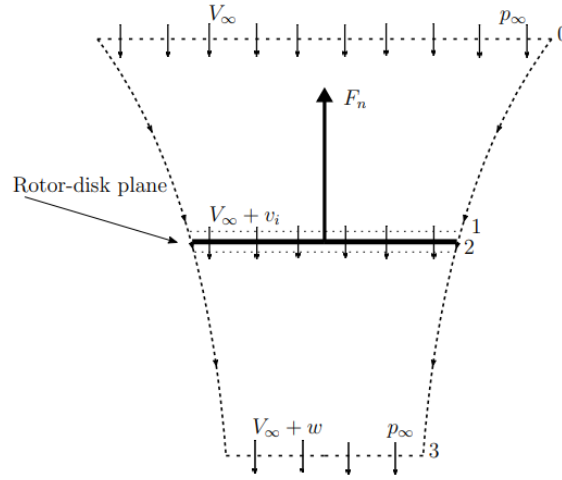


Figure 1: 1D axial momentum theory (upward flight) [3]

The 1-D axial momentum theory was employed as an initial VTOL power estimation model. This is illustrated in Figure 1. The propeller radius was initially designed to be 0.3m in radius. Therefore, in order to achieve a climb rate of higher than 1m/s, the minimum power required for each VTOL motor could be calculated using Equation 2 [3]. In order to obtain the induced velocity of the air at the rotor disk, Equation 1 was used with a free stream velocity V_∞ value of 1m/s. Therefore, combining the two equations it was initially estimated that the minimum power required for the aircraft to takeoff with VTOL propellers was 6.54kW.

$$V_{induced} = -\frac{V_\infty}{2} \pm \sqrt{\frac{V_\infty^2}{4} + \frac{W}{8\rho\pi r^2}} \quad (1)$$

$$P = 2\rho\pi r^2(V_\infty + V_{induced})^2 V_{induced} \quad (2)$$

3 Propeller Design

3.1 Preliminary Propeller Material Selection

The propellers were found to be subject to much higher bending stresses than centripetal stresses [4]. The material selection objective function was therefore to not yield under bending stresses. Therefore, using derivations from [5], the materials index was found to be $\sigma_{yield}^{2/3}/\rho_{material}$ in order to minimise mass. There also exists several material selection constraints which helped filtering the range of materials. Firstly, the water corrosion resistance of the material must be high since the aircraft would be operating above the sea. Secondly, the Young's modulus of the material must be high enough to resist deflection to prevent a loss of lift on the propeller blades.

Also, the material must have good enough fatigue strength to undertake high alternating stresses imposed by the propellers due to advancing blade effect during transitional flight. The price of the material was another consideration to the material selection of the propeller blade. As such, it was decided that Al-7068 would be used as a preliminary material for structural analysis purposes. Figure 9 was a merit index vs fatigue resistance ashby chart used to evaluate all ranges of materials. It could be seen that most of the materials were filtered out due to previously defined constraints. For example, the PA66 nylon carbon fibre composite had a higher merit index. However, due to its low fatigue strength it was not selected. Figure 10 showed that the selected material Aluminium-7068 had the highest merit index amongst all the materials that met the constraints. However, due to time constraints, the blade deflection and fatigue analysis were not carried out. This meant the material selection may still be subject to change in the next iteration. However, for the first iteration propeller design, this preliminary material was used to design the propeller for its ultimate load [4].

3.2 Propeller-Engine spool up time performance

In addition to the power requirements of the engine [15], the ability of the prop-engine to accelerate quickly to a specific rpm is also an important criteria of engine selection. This is to determine whether the propulsion system requires excess power for mission performance and controllability. Therefore, the prop-engine spool up times for both Pusher and VTOL were modelled to evaluate their response time using Equation 3. This relationship was also illustrated in Figure 2 and 3.

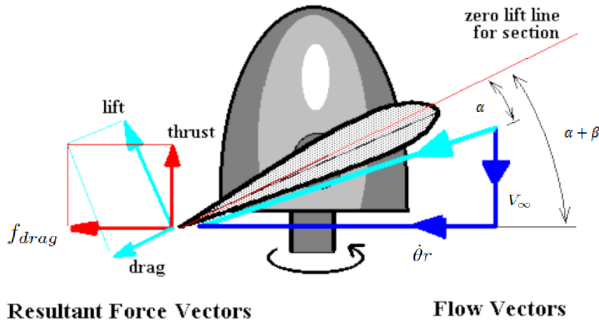


Figure 2: Propeller sectional drag schematic[6]

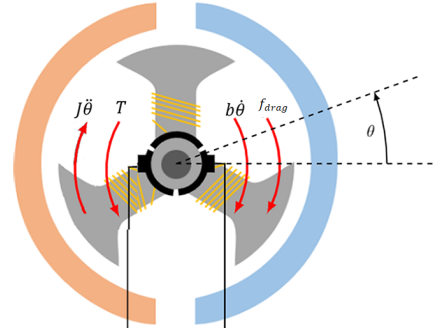


Figure 3: Spool up time equations of motion schematic[7]

$$J_{propeller}\ddot{\theta} = T - b_{friction}\dot{\theta} - f_{drag}(C_L, C_D, \alpha, V_\infty^2, \dot{\theta}^2) \quad (3)$$

The mass moment of inertia $J_{propeller}$ was estimated using the CAD model of the propeller from [2]. The prop-engine torque term T and the viscous friction term $b_{friction}$ were modelled using the manufacturer's datasheet[8][9]. $f_{drag}(C_L, C_D)$ is the propeller blade drag-induced-torque. To model $f_{drag}(C_L, C_D)$, the propeller blade was discretised into many spanwise sections. Then, the sectional drag were integrated to obtain the total drag along the propeller blade. f_{drag} is related to the propeller sectional airfoil drag and lift coefficients C_l and C_d as shown in Equation 4.

$$f_{drag}(C_L, C_D, \alpha, V_\infty^2, \dot{\theta}^2) = \int_{r_0}^{r_f} (C_l \cos(\beta + \alpha) + C_d \sin(\beta + \alpha)) \frac{1}{2} \rho c r \sqrt{(V_\infty(1+a))^2 + (\dot{\theta}r(1+a'))^2} dr \quad (4)$$

$$Re = \frac{\rho c \sqrt{V_\infty^2 + (\dot{\theta}r)^2}}{\mu}, \quad \alpha = \left(\frac{V_\infty(1+a)}{\dot{\theta}r(1+a')} \right) \quad (5)$$

$$f(C_L, C_D, \alpha, V_\infty^2, \dot{\theta}^2) = f(Re, \alpha, V_\infty^2, \dot{\theta}^2) = f(V_\infty, \dot{\theta}, V_\infty^2, \dot{\theta}^2) = f(V_\infty, \dot{\theta}) \quad (6)$$

Sectional coefficients C_l and C_d could be modelled as functions of Reynolds number Re and the angle of attack α using graphical relations derived from airfoil tools data [10] by [2]. Re and α could also be expressed in terms of the blade free stream velocity V_∞ and angular velocity $\dot{\theta}$ using Equation 5. Therefore, f_{drag} was re-modelled as Equation 6. This reduces the number of independent variables affecting f_{drag} from five to two, namely the free stream velocity V_∞ and the angular velocity $\dot{\theta}$.

Therefore, only two out of all five independent variables were left in the modelling Equation 3. A direct relationship between the propeller angular acceleration $\ddot{\theta}$, free stream velocity V_∞ and the angular velocity $\dot{\theta}$

was established as a surface plot. This was shown in Figure 11 for pusher in forward flight and Figure 12 for VTOL propeller in upward flight. These were the main flight regimes that the two propellers were design to be operate in. The angular acceleration of the motor decreases at higher rpms. This is due to an increase in prop-engine drag-induced-torque at higher tip speeds. The angular acceleration $\ddot{\theta}$ to angular velocity $\dot{\theta}$ data were then integrated to obtain the angular-velocity-time relationship of the prop-engine under full throttle input from idle speed.

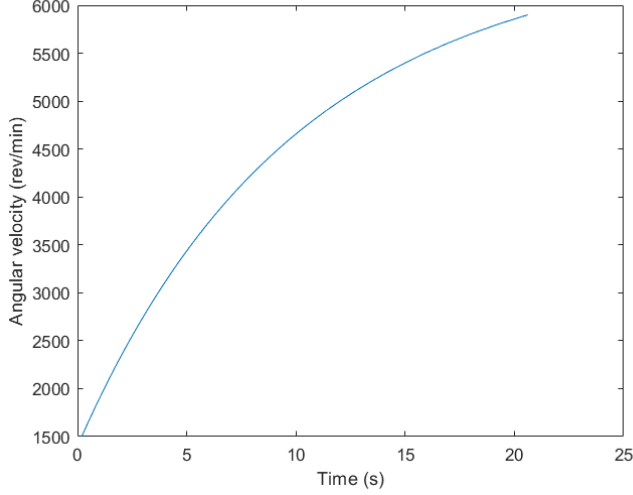


Figure 4: VTOL prop-motor min to max rpm acceleration with zero free stream velocity

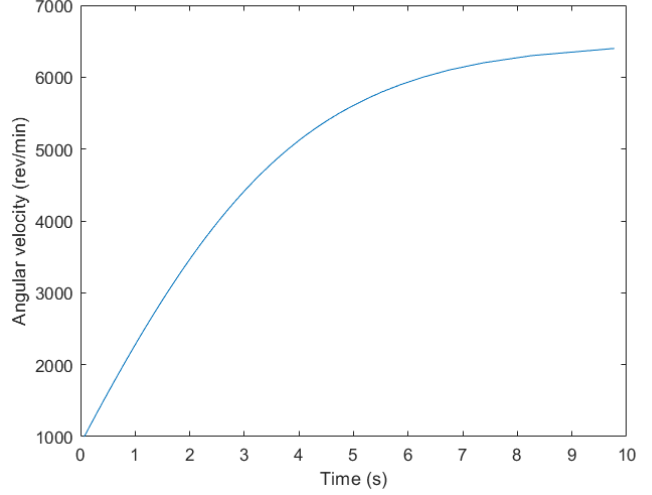


Figure 5: Pusher prop-engine min to max rpm acceleration with zero free stream velocity

On the other hand, Figure 5 and 4 shows the time required to accelerate the prop-engine from minimum to maximum rpm with zero free stream velocity. The total time taken for the VTOL and pusher prop-engine/motor to accelerate were found to be around 20s and respectively. The relationship between the free stream velocity and time to accelerate were also evaluated in Figure 6 and 7. Both propellers react much faster under their own designated flight regimes. The VTOL prop-engine accelerate much faster in upward flight when the upward free stream velocity is relatively low since it was designed to experience lower drag under low-velocity VTOL operations. On the other hand, the pusher prop-engine accelerates much faster in forward flight at higher free stream velocity due to its design adapted to fixed wing flight.

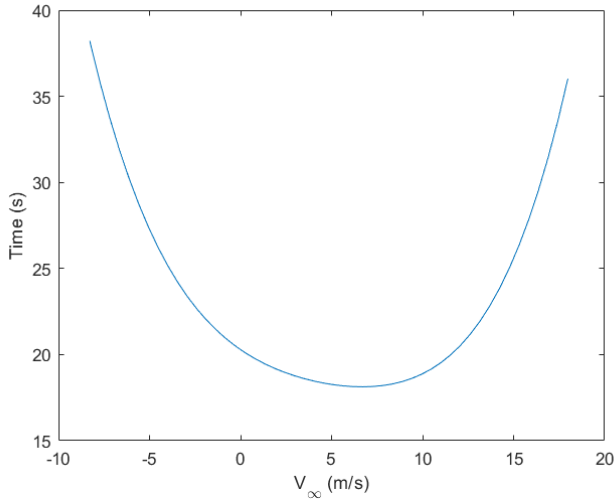


Figure 6: VTOL prop-engine acceleration time vs free stream velocity (upward flight)

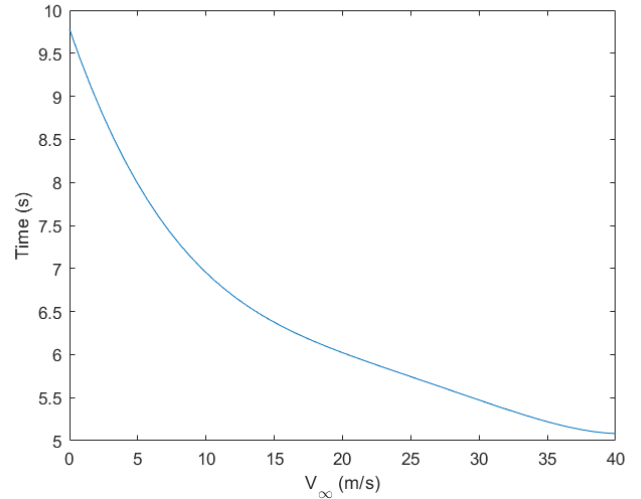


Figure 7: Pusher prop-engine acceleration time vs free stream velocity (level flight)

3.3 Validation

According to [11], the prop-engine/motor design offered enough propeller acceleration and controllability. This means no excess power was required on both the motor and engine to accelerate faster. The requirement of VTOL in 1 minute was also satisfied with a 20 seconds engine start up time[9], 20 seconds of prop-engine speed up time and 1.5 m/s VTOL speed[12]. Therefore, the motor and engine selection were finalised with considerations on performance and controllability in addition to the VTOL/flight power requirements. Further improvements on the modelling could be employed. This would be discussed in a later section.

4 Systems Design

Larger UAVs tend to have avionics specifically built for their flight applications. However, this is not the same for smaller UAVs. Smaller UAVs usually have vastly different mission profiles and performance requirements. Therefore, the range of avionics required could also vary substantially from one another. These avionics are usually not available in the industry [13]. Therefore, the avionics system design of the flying life raft were specifically designed using commercial-off-the-shelf (COTS) components that suits the specifications of the aircraft.

In this section, the design topic would be the communications system, microcontroller and the payload deployment system. On the other hand, the flight control sensors, flight computer, batteries, GPS, health and usage monitoring system, emergency landing system, actuators, ESCs and thermal camera were designed in parallel in [2][4][14][15][16].

4.1 Microcontroller

The on-board computers was designed to perform computationally expensive algorithms. This includes the human searching and recognition algorithm. It also served as the central communication station for all branches of subsystems. In addition to the flight computers, the flying life raft required additional microcontrollers (MCUs) to perform less computation intensive controls. They act as extensions from the central computer for to distribute the workload on flight control algorithms. The MCUs communicate to each other through the central computer and were designed to perform simple flight control algorithms. The advantages of using MCUs instead of additional flight computers includes lowering the weight, cost and allowing for higher compactness. This is because they are able to perform controls with self-integrated RAM and ROMs on a single chip without the need of cooling modules. Also, they provide extra I/O and powering pins for powering and communicating with the flight transducers.

There were two main considerations of MCUs for the application of our flying life raft, namely the Texas instruments MSP430 LaunchPad and the Arduino Uno Rev 3 board. Both microcontrollers of the support USB 3.0 connection to the main flight computer and work well on small UAVs according the specifications. Table 1 shows the specifications of the two MCU candidates. In term of computational specifications and the number of pins, both had similar performances. The MSP430 has a lower cost and power consumption, but did come with its trade-offs. The weight of arduino board is much lower than TI MSP430. The extra power consumption would cause an increase in battery mass. However, this effect was determined negligible since the power for both options were really low. Another criteria for microcontroller selection was the availability of online supplementary documents and supporting softwares. Arduino boards supported the use of the Arduino IDE software to directly convert and upload the flight control algorithms to the MCU. It was well documented and developed in comparison to Energia IDE. Arduino also offered extra modularity such as arduino shields to provide additional capabilities in comparison to Texas instruments. Therefore, Arduino Uno Rev3 was chosen for the application on the flying life raft.

Component	Number of I/O pins	Clock Speed (MHz), Flash Memory (KB)	Total cost (£)	Total weight (g)	Total power required (W)	Supporting software
Arduino Uno Rev 3	20	16, 32	112.2	150	1.5	Arduino IDE, open source
TI MSP430G2553	14	16, 16	65.76	594	0.525	Energia IDE

Table 1: MCU comparison[17][18]

4.2 Communications System Design

Communications system is critical for the safety of UAV operations. This is because communications allows the signalling of incoming risks and fault reports. Examples of which include bad weather signalling and air-traffic mapping. By implementing safety measures such as emergency landing prior to incoming risks, the risks of

operation could be mitigated. A remote change in mission could also be possible using the communications system.

The line-of-sight(LOS) model was employed in Figure 8 to determine the type of communications system on board. If the aircraft range is over the LOS, then the non-line-of-sight (NLOS) communication methods must be used instead. This is because the electromagnetic waves cannot travel beyond the radio horizon as shown in Figure 8. Using Equation 7 derived using geometry from [13]. During initial sizing, the flight altitude h was designed to be 400ft. Assuming the ground antenna height H to be negligible in comparison, the LOS distance R was calculated to be 24.6 nautical miles, which translates into 44.46km. It could be easily determined that a 70 km range of operation is over the LOS distance. After flight simulations were conducted, it was found that the aircraft range decreased from 70km to 54km [19]. This was still above the LOS distance. As such, SATCOM had been chosen in order to extend the range of communication by the use of low-earth-orbit satellite transmissions. Another option would be to use normal LOS communications system to surface vessels within LOS. However, it was thought to be unrealistic nor a reliable method of communications.

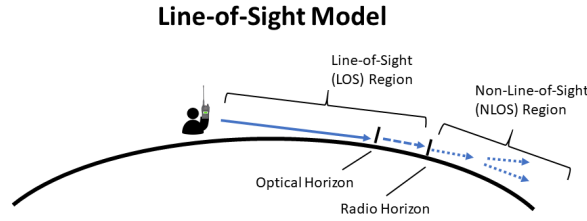


Figure 8: Line-of-sight model [20]

$$R = 1.32(\sqrt{h} + \sqrt{H}) \quad (7)$$

The SATCOM module can weight around 1-10% of the aircraft MTOW [13]. Therefore, selecting COTS would be the most accurate for performance and weight estimation. The Honeywell's UAV SATCOM system weighed around 1kg with mounted antenna[21]. This recently developed SATCOM systems offered only a slight weight penalty of 500g compared to a normal LOS system. Another advantage was that the power consumption of this system was around 7W which was found to be much lower than LOS communication systems[21]. Also, the Honeywell's UAV SATCOM system was the lightest amongst all light SATCOM systems by a at least a 30% margin. Therefore, it was chosen as the communication system of the flying life raft.

5 Systems Integration

Figure 13 showed a simplified control and monitoring systems layout for the aircraft. A more detailed layout which included wiring schematics can be found in Figure 17. As shown in Figure 13, the power is distributed from the batteries to all components through the use of wiring harness. This connects components in series and parallel to divide the voltage and current such that no DC transformers were required. The main flight computer communicate with different microcontrollers to distribute the workload and implement the flight control algorithms using the transducers connected to each microcontroller. The SATCOM, thermal camera and the GPS sensor were directly connected to the computer instead. On the other hand, the health and usage monitoring system monitors the operational condition of the entire aircraft, including the engine temperature and battery voltage. This information would be fed back to the computer to decide whether an emergency landing would be required to prevent catastrophic failure. Lastly, the EASA-anti collision light would always be turned on and flashing during operation.

5.1 Wiring Harness

The wiring harness distributes the power from the batteries to different subsystems. It allows information to be passed between different avionics systems. A detailed 2D wiring schematic were shown in Figure 17. A cleverly designed wiring harness could divide the voltage and current by connection components in series and parallel such that the power could be distributed with minimal wiring length. It could also remove the need of a DC-transformers. The wiring scheme could also help to develop an analytic estimation for the cabling weight. This would be further discussed in the weight estimation section.

5.2 Redundancies

The EASA regulation for 'specific' categories of UAVs, article 11 states that an operational risk assessment should be carried out for this category of UAV [22]. As a result, mitigation measures must be proposed for each failure mode in order to reduce the probability and minimise the impact. In order to minimise the risk probability, it was necessary to have redundancies for the critical flight control components. The inertia measuring unit, air data sensors, altimeters, microcontrollers and main computer must be duplicated. This is because there were no other components which could act as temporary replacements immediately upon failure. Duplicating the sensors could prevent catastrophic failure by introducing a Lockstep system layout [23]. In a Lockstep system, both the original and redundant set of sensors would operate in parallel to each other. This allows cross referencing, detecting and correcting errors between the systems. For the application of this aircraft, a duo modular redundancy was enough to detect the errors. This is because the aircraft mainly operates on the sea. In case of an error, the landing buoy could be deployed for immediate landing. Also, the aircraft did not require a large landing space due to its VTOL capabilities. Therefore, a tri-modular redundancy system was not required to correct errors between the measurements. This added extra safety to the aircraft operation with a weight penalty of only 520grams which was around 1% of the MTOW. This mass increase took into account both the increase in weight of the duplicated components and also the increase in power consumption, thus the battery mass. The Health and usage monitor system also was also employed to detect errors on the IC engine, batteries and motors.

On the other hand, the control surface actuators such as the aileron servo already had redundancies on the other side of the wing. A failure would only cause a decrease in the manoeuvrability thus not catastrophic. Therefore, the control surface actuators did not require extra redundancies. The IC engine may also experience failures during flight. However, the VTOL prop-motors would prevent the aircraft from falling due to stall. Therefore, the aircraft could land safely on the sea with the landing buoys. In the case of a motor failure, a normal landing could be employed at 1.3 times the stall speed with the landing buoys deployed. In most cases of failure, the mission would be aborted. However, the failure mode is unlikely neither catastrophic. In summary, the added redundancies were critical to the safety of aircraft operation and adherence to the regulations. The health and safety monitoring system also provided an extra layer of safety on fault detection as described in [4]. This ensures the lives on the sea would not be endangered by the aircraft operation.

6 Weight and Balancing

Weight and balancing was an important task necessary to determine the different inertial parameters of the aircraft such as the weight, CG and mass moment of inertia of the aircraft. This information could be very useful in developing control algorithms and the next iteration for more detailed designs.

6.1 Weight and CG Initial Estimates

The lack of UAV design methods had led to fewer options for initial weight and cg estimation methods. The methods found in literature were also described as inaccurate, due to UAVs designs being extremely volatile [13] [24]. Using parametric estimation methods and equations derived from literature review, weight initial estimates were produced based on parameters defined from initial sizing. This includes the wing aspect ratio, thickness to chord ratio, the aircraft maximum take off weight etc. The initial estimates using parametric weight estimation equations were shown in Table 2 [13][24]. UAVs in general have very different mission profiles and requirements on avionics. Therefore, the initial estimates only acted as a tool for sanity check purposes with the detail estimates.

Initial Estimation method	1. Initial sizing	2. Jay Gundlach	3. Abdulhakim Essari	Best Initial estimate
Components	Weight (g)	Weight (g)	Weight (g)	Weight (g)
Avionics	-	5150	-	5150
Payload (initially sized)	7880	7880	7880	7880
Structural - fuselage	-	10542	6642	6642
Structural - wing	-	3368	3538	3368
Structural - pylons	-	2045	1000	1000
Structural - empennage	-	1887	763	1887
Undercarriage	-	1545	1854	1545
2 stroke IC Engine for back propeller	-	3000	6800	3000
Electric propulsion system (ECS, gears etc)	-	686	-	686
4x VTOL Motors (outrunner)	-	2877	-	2877
Batteries (initially sized)	11036	11036	11036	11036
Fuel+ 2 % trapped fuel	1978	1978	1978	1978
Cooling system	-	776	-	776
VTOL + pusher Propellers	-	931	-	931
Fixed component mounts	-	2859	-	2859
cabling	-	5568	-	5568
Empty weight	32320	40458	-	35514
Total	51500	62128	-	57184

Table 2: Weight initial estimates from initial sizing [13] [24]

6.2 Aircraft Layout

6.2.1 Assumptions

Most aircraft components such as payload and avionics were found to be near-prismatic/cylindrical. Therefore, their shapes were approximated for the purpose layout design, moment of inertia (MoI) and CG estimation. The engine was large and heavy enough and therefore approximated using a more accurate T-shape instead. The density of the material were assumed to be constant throughout the approximated shapes.

For simplicity, components with weights of less than 10 g and 20 mm³ in size were ignored for both the layout and weight and balancing purposes. The accuracy of the estimation would be equally valid since the actual COTS components could have a larger deviations in manufactured weight than a total of 82g of underestimation for the aircraft. Redundancies of significantly large and heavy equipments were not ignored. This includes the microcontrollers and main flight computer.

6.2.2 Constraints and Optimisation

Most UAVs employ a slightly different layout compared to general aviation aircrafts in order to optimise for their specific design purposes. For the application of the flying life raft, there were several constraints to the layout of the aircraft each with their solutions as shown in Table 4. There were also several layout optimisation objectives defined in Table 3 each with their solutions listed. The entire layout designing process were determined with close relation to the structures and the control designs. These solutions could be subject to change in the second iteration when there are additional information available on the components, allowing rooms for further optimisations. The CAD and the drawing of the final layout of the fuselage could be found in Figure 14 and 15 with detailed annotations. The drawing of the layout of the entire aircraft could be found in Figure 16.

Optimise for	Solution
Higher Compactness	Use up as much fuselage space as possible
Better Aerodynamics: finess ratio	Stack components only below the minimum fuselage diameter
Higher Controllability	aircraft CG moved forward to provide moment arm for tail
Lower Weight	Wing and payload mounted on the same heavy frame

Table 3: Layout optimisation and solutions

Constraint	Solution
Engine must power the back pusher propeller	Engine placed at the back, aligned with the propeller centre-line
Payload deployment should not cause a loss in stability	Payload modules placed around the CG, aircraft balanced around payload
Optical sensors must have wide field of view	Thermal camera and LiDar sensor placed at the front of the fuselage
Avionics must avoid vibration sources	Avionics placed in the front, further away from the engine
Avionics must avoid electromagnetic interference	Avionics placed far away from Motors and ESCs
Components must not be impacted by rough buoy landing	Fuselage floor added with foam on top to absorb impact forces
Wing should not produce excessive pitching moments to reduce control effort	Wing aerodynamic centre placed around the CG
Ycg of the aircraft must be along the centreline of fuselage to reduce control effort	Components placed symmetrically across the fuselage centre-line
Thrust pitching moment arm must be small to reduce control effort	Place components such that the propeller centre-line is close to Zcg
The aircraft must be statically stable (longitudinal)	Iterative layout design with the control team, enough tail size and moment arm from CG
The aircraft must be Dynamically stable (longitudinal)	Iterative layout design with the control team, $X_{cg} \text{ less than } 0.66m$ [25]
The aircraft must be statically stable (lateral)	Iterative layout design with the control team, control surfaces far enough outboard[26]
The aircraft must be Dynamically stable (lateral)	Iterative layout design with the control team
Undercarriage must be placed for landing stability and avoid tail strike during landing	Place undercarriage such that tip back angle does not exceed 15 degrees
The VTOL propellers must not be blocked away from free stream excessively	VTOL propellers placed above the motor and wing.
VTOL propellers must not interact with other components excessively	VTOL placed to have at least 5cm clearance from all working parts
The payload must be deployable with the layout	No extra components was placed underneath the payload to clear the bottom for deployment

Table 4: Layout constraints and solutions

6.3 Weight and CG Detailed Estimates

The detailed aircraft power and flight control systems weight estimation was mainly based on COTS components selected by various system designers[2][4][14][15]. The detailed structural weight estimations were based on the analytical structural weight estimations [30][29][28][27]. The weights and dimensions were derived from these designs for layout and weight estimation purposes.

6.3.1 Assumptions

As mentioned previously, components were approximated as prismatic or cylindrical shapes for CG and moment of inertia estimation purposes. However, this did not include heavy structural components such as the wing, tail and fuselage. Initially the idea was to retrieve CG and moment of inertia information from the structural CAD design for higher accuracy. However, due to time constraints, the CAD components were only partially complete without the heavy frames. Therefore, approximations were made on the CG and the moment of inertia of the structural components.

This estimation method was based on the geometric parameters of the structural components. For example, the wing CG was estimated using a trapezoidal pyramid approximation. Using the taper ratio λ , aspect ratio AR and the wing area S_w , the centroid of the rectangular pyramid wing could be found. However, in order to capture the airfoil shape, the chordwise centroid position was modified using the centroid of the wing's airfoil section. The centroid of the wing's airfoil section was found by using the MATLAB functions in the appendix. The centroid of the wing was found to be 42.6% of the chord at 44.4% of the semi-span of the wing using trapezoidal pyramid approximations. Table 5 shows the approximations in the structural components of the aircraft. Components that were very small and light were assumed to be negligible similar to the layout design.

Components	X centroid	Y centroid	MoI shape approximation
Wing	Wing mounted position +42.1% chord at 44.4% semi-span	44.4% semi-span	trapezoidal pyramid
Horizontal tail	Horizontal tail mounted position + 41.4 % chord at 44.4% semi-span	44.4% semi-span	trapezoidal pyramid
Vertical tail	Vertical tail mounted position + 42 % chord at 44.4% semi-span	end of Horizontal tail + half the thickness of airfoil	trapezoidal pyramid
Fuselage	47% of the CG due to after body tapering, pusher configuration [31]	at the fuselage centre-line	cylindrical
Pylons	Wing mounted position + Mean aerodynamic chord at 80% of wing $\pm$ half the length of pylon	pylons Placed at 80% of wing span	cylindrical

Table 5: CG estimation and Moment of inertia shape approximations according to aerodynamic shape

6.3.2 Methodology

The weights for different components were estimated using a mix of three different methods. The first method involved using the manufacturer's datasheet for the COTS components that suit the requirements of the flying life raft. Most of the aircraft components were estimated using this method. The second method involved using CAD or analytical weight calculations of the components specifically designed for this flying life raft. This includes the batteries, fuselage, wing, tailplane and the pylons [4][27][28][29][30]. These two methods were the most accurate methods available at this design stage. The third method involved the use of parametric weight estimation relations (PWERs). This method utilises the parameters such as the aspect ratio AR and the wing area S_w defined from initial sizing to adjust the estimation for different components. However, this method would produce result with a lower fidelity. Therefore, the usage of the first and second method were maximised where possible. In order to estimate the CG and moment of inertia, the layout was combined with the weight estimations.

For the aircraft CG estimation, the weight estimations were combined with the layout of the aircraft to find out the total moment contribution of each component on all three x, y, and z directions. This was then divided by the total weight of the aircraft to find out the average CG of all components. This process was illustrated in equation 8 [13]. The results for the aircraft overall CG estimation was shown in Table 8.

$$X_{CG} = \frac{\sum W(i)x(i)}{W_{Tot}}, Y_{CG} = \frac{\sum W(i)y(i)}{W_{Tot}}, Z_{CG} = \frac{\sum W(i)z(i)}{W_{Tot}} \quad (8)$$

The moment of inertia could then be estimated. For complex geometries, CAD or approximated geometries were used to calculate the moment of inertia. The two main approximations were prismatic and cylindrical shapes. The self moment of inertia for these shapes were evaluated using the equations derived in [32]. Therefore, a complete list of the self-moment of inertia of each component could be found in Figure 13. Using the parallel axis theorem as illustrated in equation 9, 10, 11 and 12, the global moment of inertia contribution of each component could be found. In the end, all contributions were summed up to obtain the overall aircraft moment of inertia. This was shown in Figure 7.

$$I_{x0} = \sum W(i)y(i)^2 + \sum W(i)z(i)^2 + \sum I_x(i) - W_{Tot}(Y_{CG}^2 + Z_{CG}^2) \quad (9)$$

$$I_{y0} = \sum W(i)x(i)^2 + \sum W(i)z(i)^2 + \sum I_y(i) - W_{Tot}(X_{CG}^2 + Z_{CG}^2) \quad (10)$$

$$I_{z0} = \sum W(i)x(i)^2 + \sum W(i)y(i)^2 + \sum I_z(i) - W_{Tot}(X_{CG}^2 + Y_{CG}^2) \quad (11)$$

$$I_{xz0} = \sum W(i)x(i)z(i) - W_{Tot}X_{CG}Z_{CG} \quad (12)$$

6.3.3 Cabling and Mounting Weights Estimation

Cables had a much more generic layout and thus it was not possible to approximate cables into any shapes, nor it was to accurately estimate their weights. Although a detailed wiring harness design had been produced in Figure 17, it was a 2D schematic. It did not show the extra distance the wires would need to travel around bulky components. Therefore, a length times wire density quasi-analytical approach was not employed. The increase in accuracy of the estimation would not be increased by much. Given the time constraints, it was more reasonable to design the systems in more detail to make sure the wiring schematic would not change in the future iterations. Therefore, a parametric cabling weight estimation was used instead as shown in Equation 13

with f_{wiring} taken to be 0.25 for small UAVs. However, the quasi-analytical approach was partially calculated to ensure the parametric weight estimation was accurate in terms of magnitude.

$$W_{wiring} = f_{wiring}(\Sigma W_{avionics} + \Sigma W_{payload} + \Sigma W_{comms}) \quad (13)$$

Most avionics were placed in the front with short cable lengths and small power requirements. The motors were placed much further away from other avionics due to interference and propelling reasons. It also required much more power and therefore current through the powering wires. This requires a more demanding gauge wire such as the AWS gauge-3 cable rated for 70A application. It was found to be almost 10 times as heavy as any other wiring used in the system. Therefore, the weight of the motor wirings would represent the majority of the wiring weights. Combining the systems layout Figure 13, the aircraft fuselage layout Figure 15 and structural layout Figure 16 gives the minimum length of wiring required for the motor. This gives a minimum cable length of 5.66m. The AWS gauge-3 cable has a density of 0.287 kg/m [33]. Therefore, the total weight of the motor wiring was found to be 1.6kg. This was thought to be comparable with the current cable weight estimation. Therefore, the parametric estimation was valid for an initial cable weight estimation. However, any cabling CG and moment of inertia estimations would simply be inaccurate at this stage. Therefore, the effects of cabling on the CG and moment of inertia were neglected for the first iteration. The CG of all cables were assumed to be the CG of all avionics.

On the other hand, mounting mechanisms were yet to be designed. This required accurate information about the components and the structural design of the aircraft. Therefore, parametric weight estimation was used at this design stage. In this case, the CG of the mounts were estimated to be the same as the CG of the components. Therefore, it was added on top of the weight of the components at the final stage of weight estimation. Equation 14 was used to determine the mounting weight, with a f_{mount} of 0.1 for avionics and payload and a f_{mount} of 0.2 for engine [13].

$$W_{comp,mount} = f_{mounts}(W_{comp}) \quad (14)$$

6.3.4 Results

The detailed weights, CG and moment of inertia estimates for different groups of components were shown in Table 6, 8 and 7. A complete expanded version for the components used, weight estimations, position estimations, CG contributions, self and global mass moment of inertia contributions of each components could be found in Table 9,10,11, 12, 13 and 14 respectively. The final MTOW of the aircraft was found to be 49.56kg. This was found to be approximately 2 kg lighter than that of initial sizing and kg lighter than initial parametric estimation from literature. On the other hand, the control team had confirmed that all the CG and moment of inertia values obtained using the current layout design were producing stabilising results for the aircraft [11][12][25][26]. Therefore, this information was finalised for this iteration and fed into the control team for further stability and performance analysis .

-	COTS components	Analytic	Parametric	Total	% difference from best initial estimate
Components	Weight excluding mounts (g)	Weight (g)	Weight (g)	Weight including mounts (g)	%
Avionics	3670	-	-	4099.75	-25.6
Payload	7880	-	-	8668	10.0
Structural - fuselage	-	3740	-	3740	-43.7
Structural - wing	-	7480	-	7480	122.1
Structural - pylons	-	714	-	714	-28.6
Structural - tailplane	-	1819	-	1819	-3.6
Undercarriage	1000	-	-	1100	-28.8
2 stroke IC Engine for back propeller	2895	-	-	3474	15.8
ESC	312	-	-	343.2	-50.0
4x VTOL Motors (outrunner)	2520	-	-	2772	-3.7
Batteries	6200	-	-	6820	-38.2
Fuel+ 2 % trapped fuel	4590	-	-	5049	155.3
Cooling vents	-	0	-	0	-
VTOL + pusher Propellers	943	-	-	943	1.2
cabling (UAS design)	-	-	2541	2541	-11.1
Fixed component mounts	-	-	3196	-	-
Total	30009	13753	5737	49563	-13.3

Table 6: Detailed component weight estimations

Quantity	Value (g mm ²)
Ixx	6870000000
Iyy	5479000000
Izz	10840000000
Ixy	0
Ixz	-211500000

Table 7: Detailed aircraft mass moment of inertia estimations

Quantity	Value (mm)	%aft of fuselage
Xcg	614.2	0.460
Extreme case 1: Xcg after payload drop	616.3	0.462
Extreme case 2: Xcg empty fuel tank	593.7	0.445
Landing buoy 1 cg	405.0	0.300
Landing buoy 2 cg	927.2	0.687
Ycg from pusher centre line	0.0	N/A
Zcg from pusher centre line	-3.5	N/A
Extreme case 1: Zcg after payload drop	-13.9	N/A
Extreme case 2: Zcg empty fuel tank	-3.5	N/A
Length of fuselage	1350.0	N/A
Diameter of fuselage	352.0	N/A
MTOW	49563	N/A

Table 8: Detailed aircraft CG estimations

7 Discussions and Further Improvements

In material selection, AL-7068 was selected as a preliminary material for structural analysis purposes [4]. This is subject to change in the next iteration. However, the aircraft was designed to be capable of at least 100 cycles of flight [34]. It was found that the biggest source of alternating load was on the VTOL propellers during transitional flight. This source of load could be used to estimate the fatigue strength required for validating the material selection. Aluminium-7055 and Stainless steel MS-42 were thought to be the two primary competing candidates with slightly low merit indices but higher fatigue strengths.

Slightly higher merit index Not only lower fatigue strength, also higher scatter knockdown factor which makes it an unreliable choice for our application of life saving. Therefore, the realistical merit index depends on the accuracy of the manufacturing methods. The cost of CFRPs are also much more expensive than aluminium.

For prop-engine spool up time modelling, it would add extra accuracy to evaluate the VTOL prop-engine rpm-time performance for transitional flight. However, this requires more time to model due to VTOL props being subjected to a larger range of free-stream velocities at varying angle of attacks. This also requires considerations on the advancing blade theory. Due to time constraints, the upward flight model was employed for VTOL prop-motor transitional performance estimation. However, it is of reasonable accuracy at this design stage to model VTOL prop-motor transitional flight performance with the upward flight acceleration performance.

In the two flight regimes, the propeller would experience a similar free-stream velocity to rotational velocity ratio already evaluated. This is because the aircraft transitional flight was not designed to have rapid rises and drops in the aircraft up-right position [12]. Therefore, the range of angle of attacks the two models cover would be similar. On the other hand, the variation in airspeed and therefore Reynolds number on the propeller blades would also have a minimal effect because the flow is far beyond turbulence transition point. The aircraft propeller blade is also rotating at a much higher velocity than the stall speed of the aircraft during transition flight. Therefore, the advancing blade effect would not cause major discrepancies between the two models. Therefore, using equation 6, the drag-induced torque f_{drag} would be largely within the spectra already evaluated for VTOL prop-engine. Also, the drag induced by the propeller blade is largely dependent on the rotational velocity of the tip of the blade. It is of much larger magnitude than the free-stream velocity below stall speed. Therefore, the VTOL prop-engine upward flight acceleration model can approximate transitional flight reasonably well at this early design stage. Since the aircraft is subject to change at a more detailed stage, more efforts were put into systems designs in this iteration. However, transitional flight spool up model could be an improvement on the modelling accuracy for the next iteration.

The detailed weight and balancing estimation showed a 2kg decrease in MTOW compared to the initial sizing stage. It showed a 7.6kg decrease in MTOW compared to the best initial estimates. This is as expected due to the lack of accurate initial sizing and weight estimation methods present for UAV design. There were

big discrepancies in weight estimations for the batteries and fuel weight ratios and the wing weight as shown in table 2. This was due to a 38.2% overestimation of battery weight in the initial sizing stage and an 155% underestimation on the specific fuel consumption of 2 stroke piston engine according to [4][15]. There was also a 122% underestimate on the wing weight and a 43% over estimate on the fuselage weight [28] [30]. Although the final weights were found to be similar, the detailed estimate was much more accurate and well thought-out in comparison to the initial estimates and initial sizing weights.

The 2.5kg extra allowance on MTOW allows for additional payloads. Additional fuel or modular payloads could be employed on the aircraft as discussed in [35]. As discussed previously, the simulations showed a reduction in range by 23% due to the inaccurate estimation of the drag coefficient C_{D0} . Therefore, the extra 2kg could be allocated to extra fuel to make up for the range loss in range for the next iteration. However, the fuel tank would require slight modifications to allocate extra space for the fuel. Also, extra redundancies could be added to the aircraft to increase the safety of operation. The weight estimation at this early design stage achieved a weight maturity score of around 0.5 according to [13] in figure 18. This is because most of the weight and balancing estimates were carried out using Commercial-off-the-shelf components. On the other hand, components such as the wing, fuselage and batteries have gone through a more detailed design stage. The weight estimation resulting from these designs are analytical and suits the requirements of our aircraft.

However, miscellaneous weights such as the cabling and the mounting weights presents difficulty for an accurate estimation. It would be relatively difficult at this design stage to obtain an accurate design of the mounting methods. It was relatively difficult to obtain a complete cabling schemes for each individual components at this stage due to the time constraints. These estimations have been kept parametric for this iteration. However, these estimations have been sanity checked with calculations to increase fidelity. The cabling weight in particular also posed a small amount of inaccuracies on the cg and mass moment of inertia calculations. However, cabling weights shared below 5% of the MTOW. Therefore, the current estimation was more than accurate enough to be fed to the control team for the initial flight controller development.

In terms of weight and cg estimation accuracy, several improvements could be made in the next iteration. The cabling and mounts weight estimations had a lower maturity score. Therefore, they would need to be prioritised in the next iteration. The cabling could use the aircraft layout for a better weight estimation. The component mounts on the other hand, could be designed since the fuselage design was complete in this iteration. On the other hand, the COTS components could increase their maturity score by purchasing sample products from the manufacturer and measure the different inertial properties directly. Also, previously assumed negligible components could be added back to the weight and balancing calculations to reduce plant mismatch for the control team. Another option would be to develop a more accurate CAD of the aircraft layout and structural design. This would increase the accuracy of the cg and moment of inertia estimation. This is because the current assumption was to approximate components inside the fuselage as prismatic or cylindrical shapes with constant density.

In future iterations, there are a few issues to be addressed for the entire group. Regarding the aerodynamic analysis of the aircraft [36], a more accurate estimation of C_{D0} could be used to design the aircraft in the next iteration instead. This could further optimise the aircraft for the specified mission performance [34]. Overall, the resulting flying life raft product was accurate and successful. The performance could be further be improved with a few design changes in the future.

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Appendix

7.1 Ashby Charts

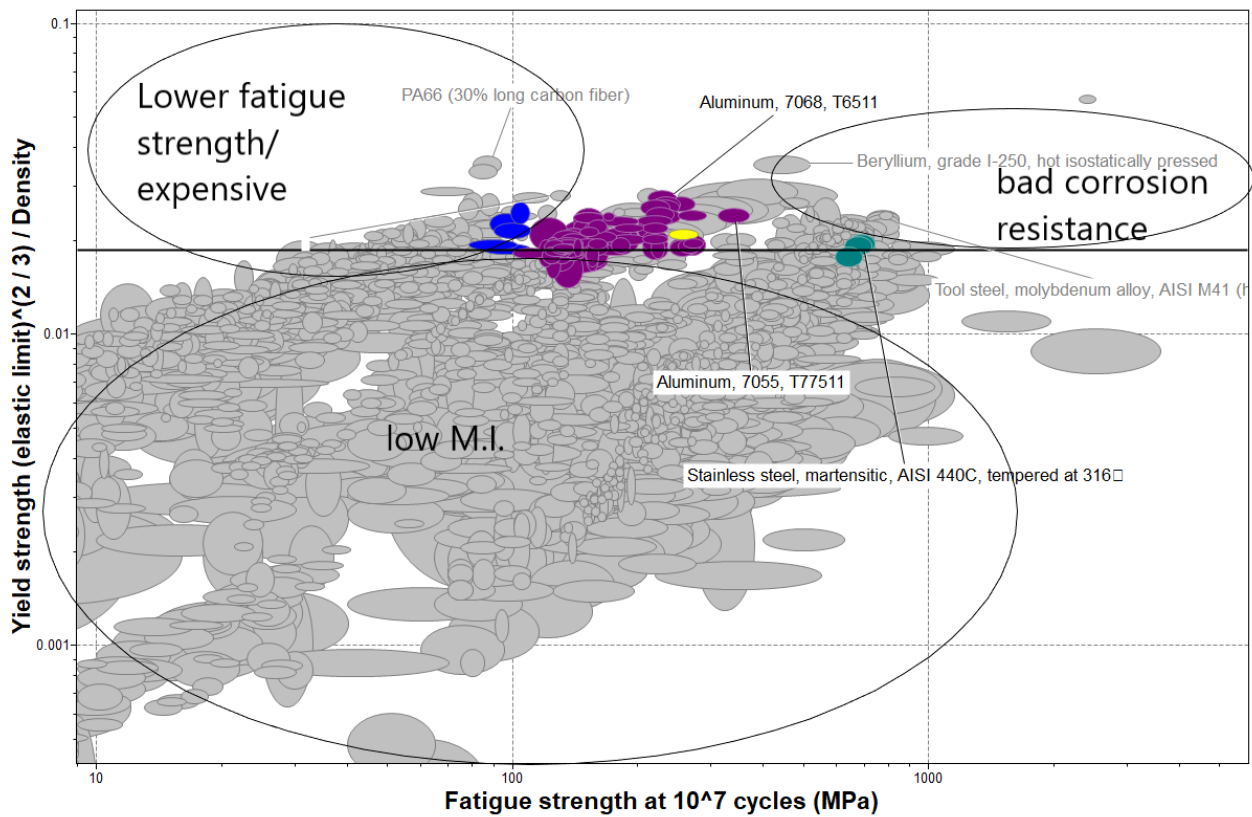


Figure 9: Ashby Chart for propeller blade material selection: merit index vs fatigue

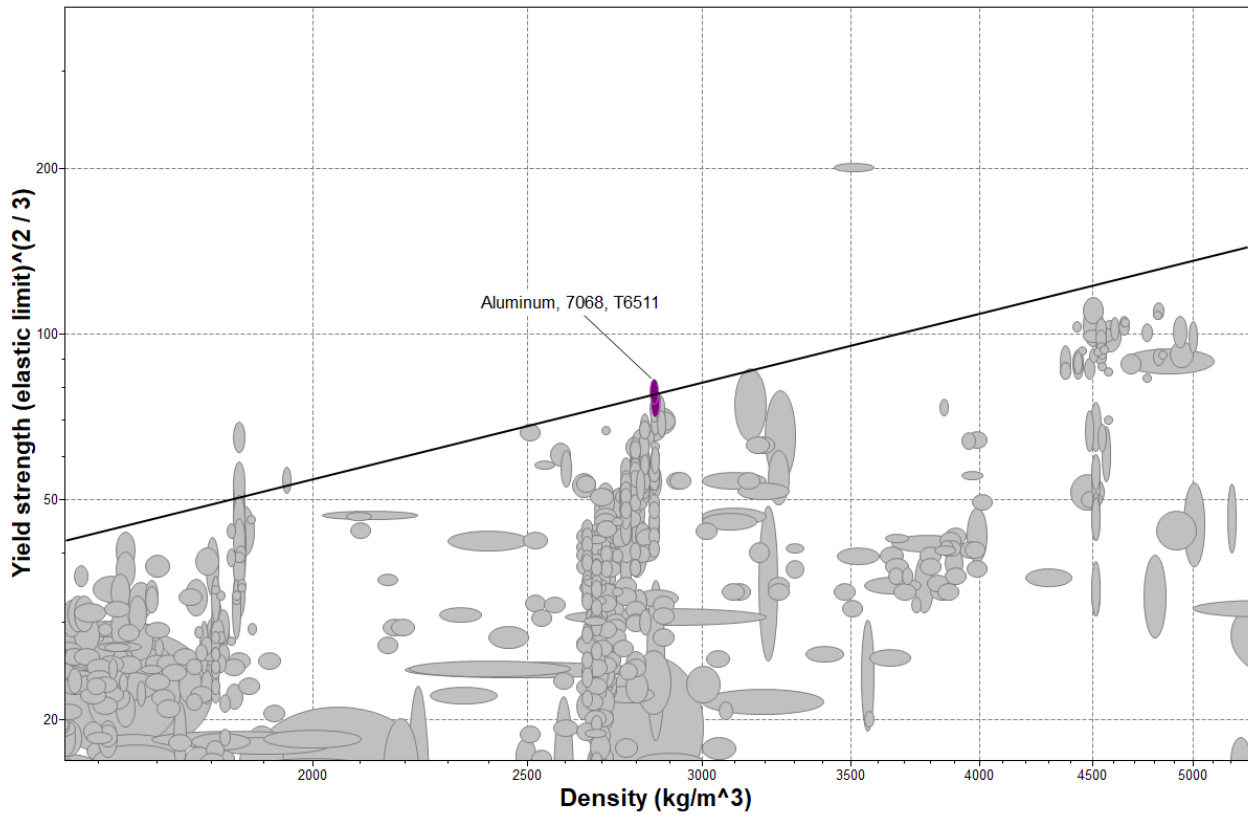


Figure 10: Ashby Chart for propeller blade material selection: merit index plot

7.2 Pusher/VTOL prop-engine acceleration vs free stream velocity vs rpm

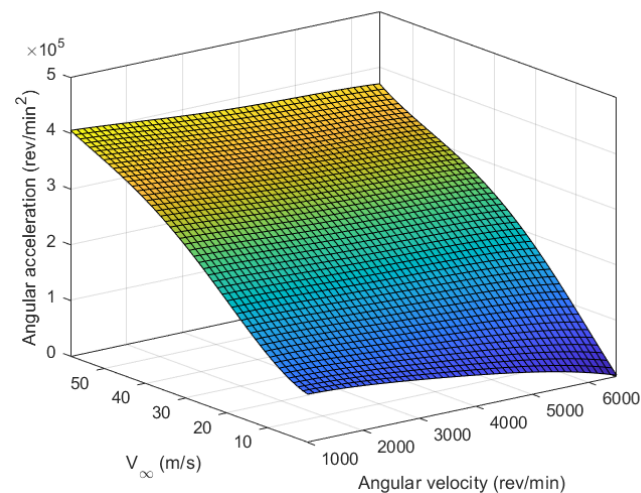


Figure 11: Pusher prop-engine acceleration vs free stream velocity vs rpm

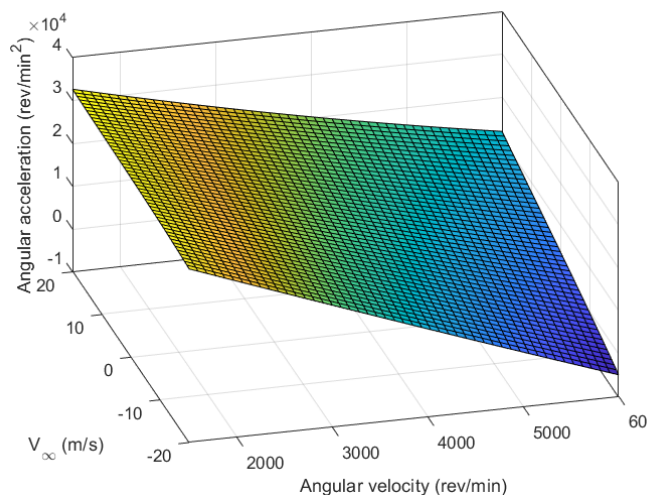


Figure 12: VTOL prop-motor acceleration vs free stream velocity vs rpm

7.3 Final Systems Layout

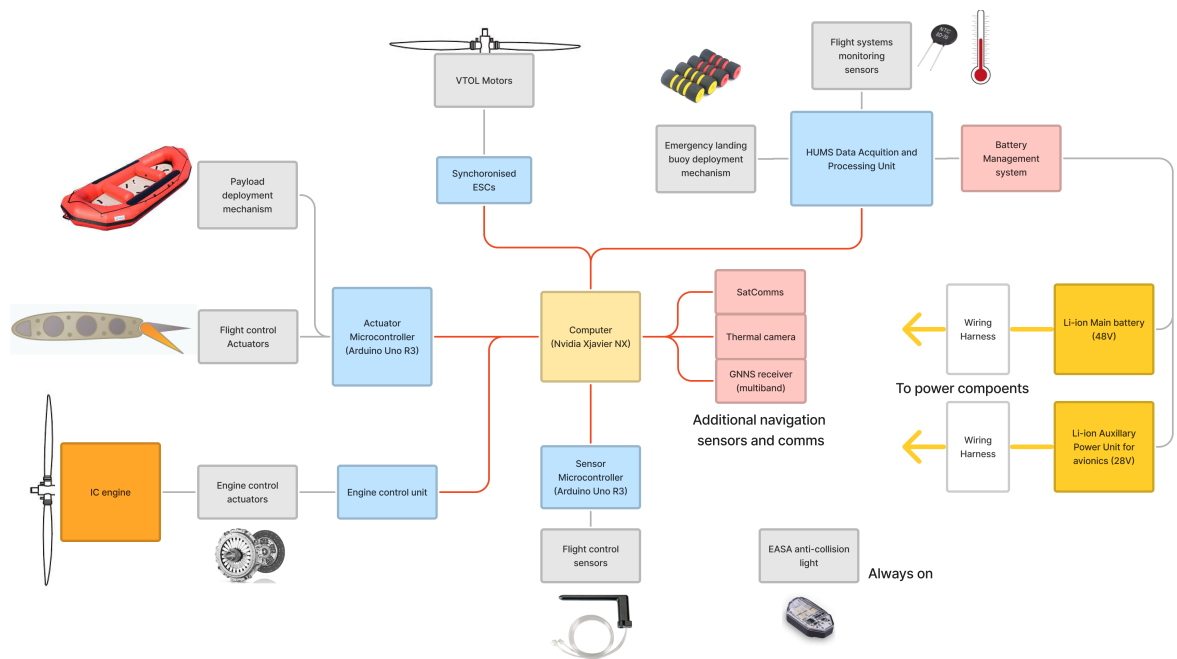


Figure 13: Systems layout

7.4 Layout CAD

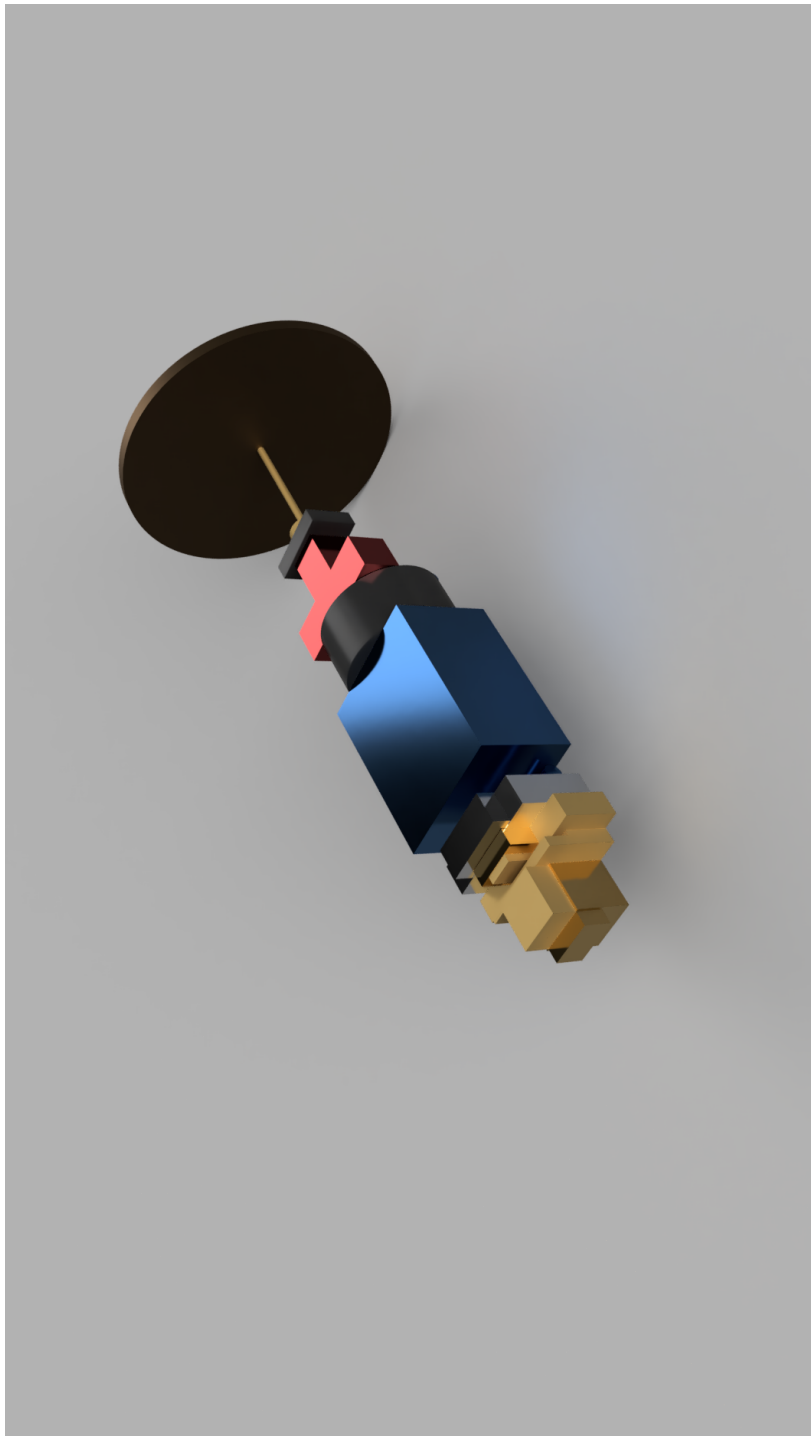


Figure 14: Layout CAD with Prismatic/Cylindrical approximations

7.5 Layout Drawing

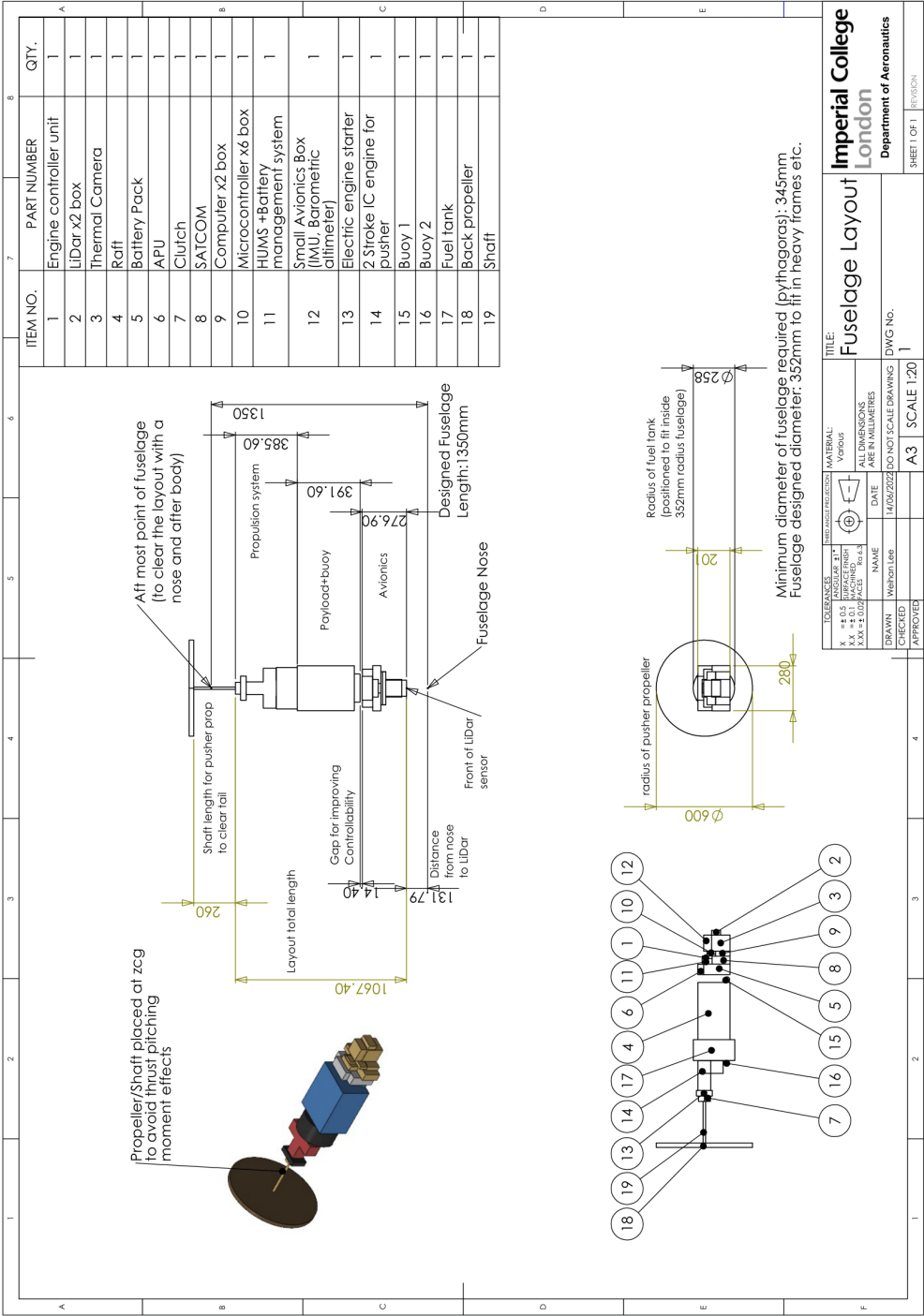


Figure 15: Layout CAD Drawing with Prismatic/Cylindrical approximations

7.6 Aircraft Structural Design Drawing

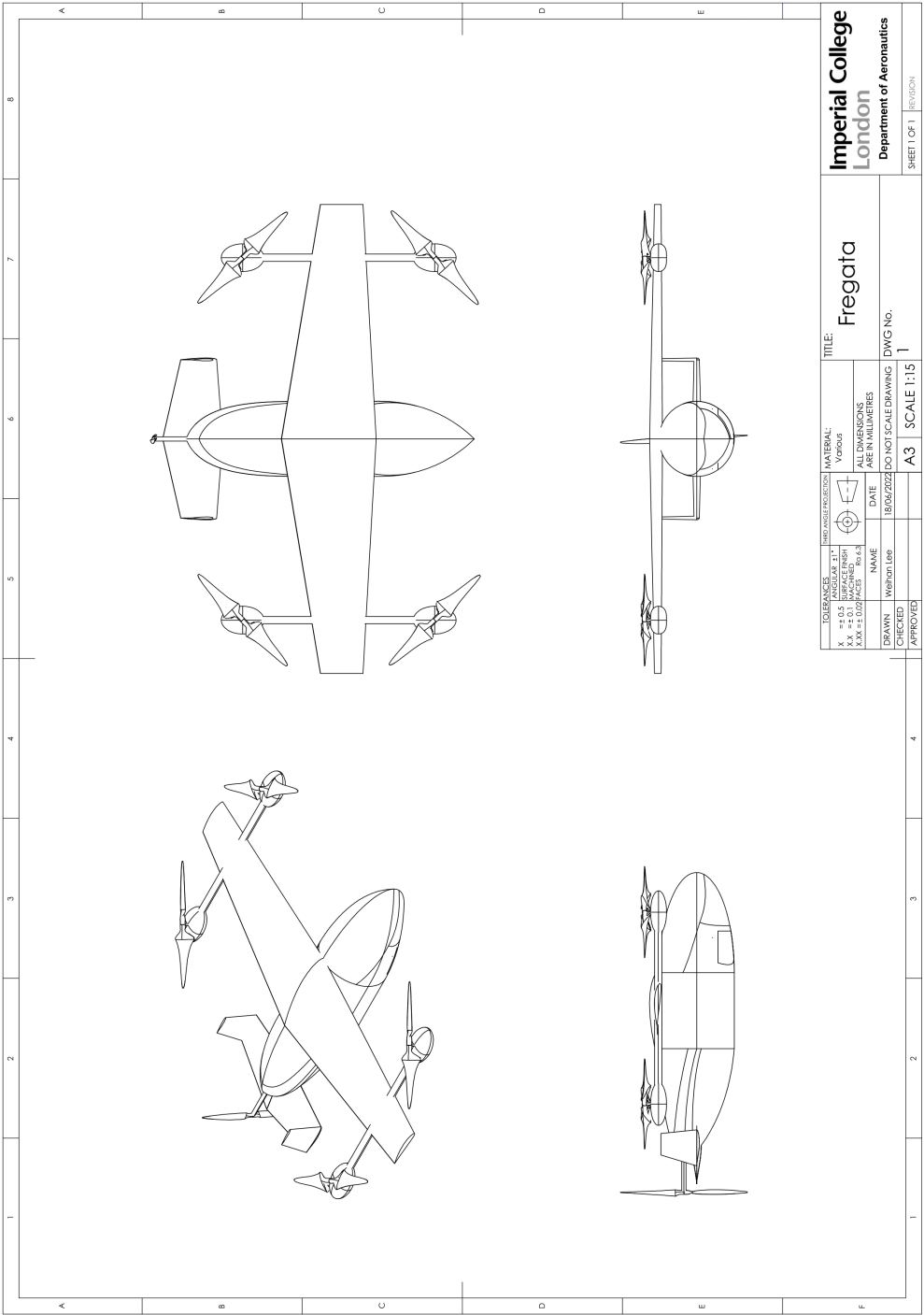


Figure 16: Aircraft Structural Design Drawing from Structures CAD [27]

7.7 Detailed Wiring Schematic

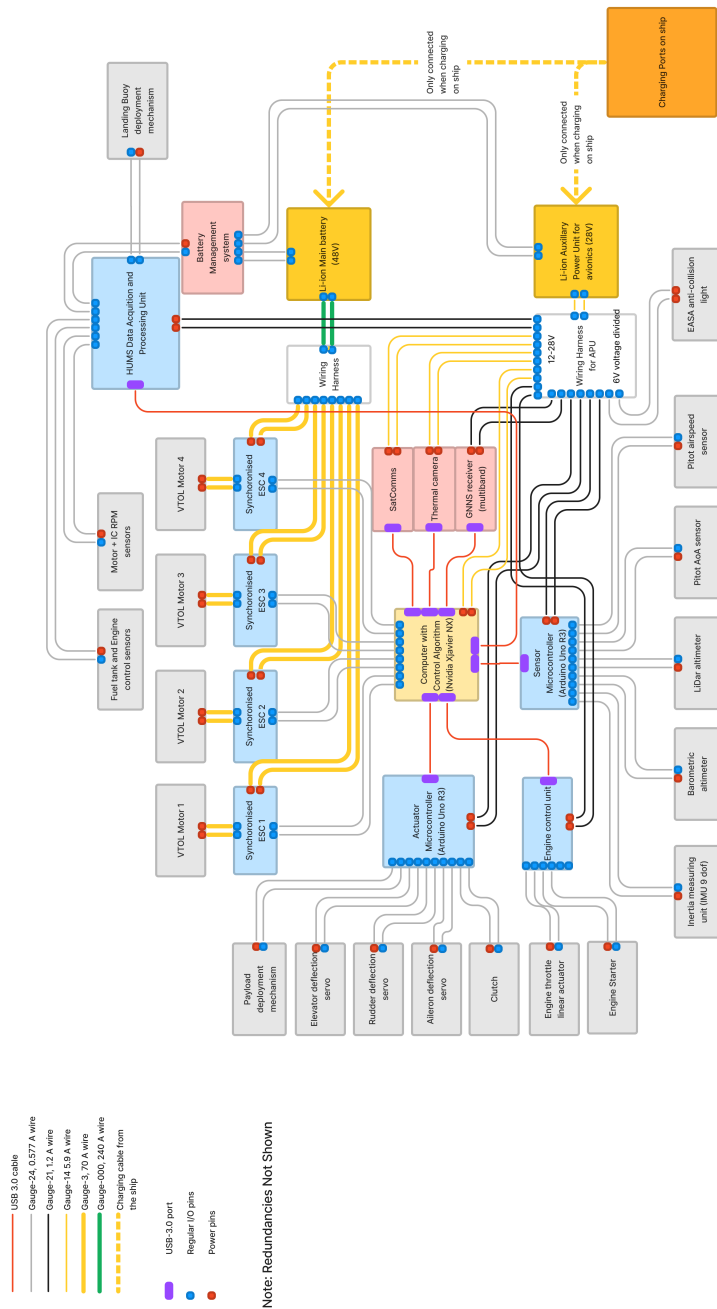


Figure 17: Detailed Wiring Schematic

7.8 Detailed weight, cg and inertia estimation breakdown

Notes:

1. Significant figures only rounded at the last stage of estimation within main body of the report for accuracy.
2. Excel lacks fast significant figure rounding capabilities for large number of datasets so doing so manually would be difficult given the time constraints.
3. Most components were picked from COTS parts.
4. Components were approximated to have cylindrical/prismatic shapes with constant densities.

Components	Weight + Mounting weight (g)	Name of product used
Avionics - basics sum		
Computer for flight control and processing	378.4	NVIDIA JETSON XAVIER NX MODULE 8GB with heat sink
Sat comm	426.8	Honeywell's small UAV SATCOM
Microcontroller	165.0	Arduino Uno Rev 3
HUMS+Battery Management System	330.0	BMS TYVA
EASA anti-collision green light	30.8	DR-02 Strobe Drone Light
Avionics - sensors sum		
Barometric Altimeter	6.6	adafruit bmp388
Engine Controller unit (ECU)	93.5	Currawong engineering ECU
Thermal sensor	11.0	Cylinder Head Temperature Sensor
Rpm sensor	27.0	SBS-01R0
Inertia measuring unit (IMU)	27.5	ORIENTUS MEMS AHRS
Pitot airspeed sensor	55.0	MATEK SYSTEMS ASPD-4525
Lidar Altimeter	77.0	lightware SF11/C
AOA sensor	220.0	MATEK SYSTEMS ASPD-4525
GNSS Receiver (multi-band)	negligible	ZED-F9T module
Thermal Camera	495.0	GC3-T 4K 3-Axis Gimbal Thermal Camera
Avionics - actuators sum		
Aileron servo	134.2	Futaba S9155
Rudder servo	37.4	Futaba S3114
Elevator servo	74.8	Futaba S3016
Payload mechanism servo	74.8	Futaba S3016
U/C sum		
Landing Buoy	1100.0	Designed by structures
Payload sum		
Raft	8668.0	Initial sizing: Aero Compact 4 Man Liferaft
Structural - fuselage sum		
Aluminium Structures + impact absorption foam floor	3740.0	Designed by structures: Aluminium 7055 T74511
Structural - wing sum		
Aluminium	7480.0	Designed by structures: Aluminium 7055 T74511
Propulsion sum		
2 stroke IC Engine for back propeller	3474.0	DLE 120
ESC VTOL	343.2	KDE-UAS95HVC
Motors (brushless)	2772.0	KDE6815XF-205
VTOL prop	884.3	Aluminium 7068
Pusher prop	58.4	Aluminium 7068
Battery Pack	6050.0	Li-ion
APU	770.0	Li-ion
Electric Engine starter (motor+frame)	778.0	Electric auto starter + Brushless Outrunner Motor
Clutch	506.0	Electromagnetic Clutch, DC 24V, 200Nm/400Nm
Shaft	63.0	Aluminium 7055
Throttle actuator	88.0	Actuonix Miniature Electric Linear Actuator
Fuel+trapped	5049.0	gasoline
Structural - pylons sum		
Aluminium x 4	714.4	Designed by structures: Epoxy/HS Carbon fiber
Structural - H tailplane sum		
Horizontal Aluminium	1582.8	Designed by structures: Aluminium 7055 T74511
Vertical Aluminium	236.0	Designed by structures: Aluminium 7055 T74511
Cable sum	2541.3	(AWS Gauge 000, 3, 14, 21, 24) & (USB 3.0)
Total sum	49563.1	
Mounting weight sum	3196.2	Mounting Plates, screws and bolts

Table 9: Detail components break down

Components	Weight+mounting weight (g)	Length (mm)	Width (mm)	Height/diameter (mm)
Avionics - basics				
Computer for flight control and processing X2 box	378.4	31	206	90.5
Sat comm	426.8	50	281.9	111.8
Microcontroller (python)X6 box	165	20	205.8	53.4
HUMS+Battery Management System	330	35	110	62
EASA anti-collision green light x2 box	negligible	negligible	negligible	negligible
Avionics - sensors				
Barometric Altimeter x2 box [inside Small avionics box (SAB)]	negligible	negligible	negligible	negligible
Engine Controller unit (ECU)	93.5	16	61	60
Thermal sensor X2 (engine mounted)	11	negligible	negligible	negligible
Rpm sensor X5 (motor mounted)	negligible	negligible	negligible	negligible
Inertia measuring unit (IMU) x2 box (inside SAB)	negligible	negligible	negligible	negligible
Pitot airspeed sensor box (wing mounted)	55	negligible	negligible	negligible
Lidar Altimeter x2 box	77	30	100	56.5
AOA sensor (Pressure differential 4 pitot) x2 (wing mounted)	220	negligible	negligible	negligible
GNSS Receiver (multi-band) x2 box (inside SAB)	negligible	negligible	negligible	negligible
Thermal Camera	495	100	110	115
Avionics - actuators				
East Aileron servo1	67.1	40.0	36.6	20.0
West Aileron servo2	67.1	40.0	36.6	20.0
East Rudder servo 1	37.4	31.0	16.0	30.7
West Rudder servo 2	37.4	31.0	16.0	30.7
East Elevator servo 1	37.4	31.0	30.7	16.0
West Elevator servo 2	37.4	31.0	30.7	16.0
East Payload mechanism servo 1	37.4	30.7	31.0	16.0
West Payload mechanism servo 2	37.4	30.7	31.0	16.0
U/C				
Landing Buoy front	550.0	176.0	N/A	35.6
Landing Buoy back	550.0	176.0	N/A	35.6
Payload				
Raft	8668.0	356.0	280.0	200.0
Structural - fuselage				
Aluminium Structures + impact absorption foam floor	3740.0	1300.0	354.0	352.0
Structural - wing				
East	3740.0	347.7	1100.0	45.2
West	3740.0	347.7	1100.0	45.2
Propulsion				
2 stroke IC Engine for back propeller (T-shaped)	3474	186	265	141
ESC VTOL 1	85.8	82	37	10
Motors (brushless) 1	693	77.2	77.2	49.5
VTOL prop 1	221.07	300	300	40
ESC VTOL 2	85.8	82	37	10
Motors (brushless) 2	693	77.2	77.2	49.5
VTOL prop 2	221.07	300	300	40
ESC VTOL 3	85.8	82	37	10
Motors (brushless) 3	693	77.2	77.2	49.5
VTOL prop 3	221.07	300	300	40
ESC VTOL 4	85.8	82	37	10
Motors (brushless) 4	693	77.2	77.2	49.5
VTOL prop 4	221.07	300	300	40
Pusher prop	58.4	28	300	300
Battery Pack	6050	65	236.9	164
APU	770	65	148	37
Electric engine starter mounted aft of ICE	778	35	150	100
Clutch	506	33	N/A	74
Shaft	63	293	N/A	35
Throttle Linear actuator	88	negligible	negligible	negligible
Fuel+trapped Mount	5049	130.6	N/A	258
Structural - Pylons				
East-Front	178.6	320	N/A	31
West-Front	178.6	320	N/A	31
East-back	178.6	320	N/A	31
West-back	178.6	320	N/A	31
Structural - H-tailplane				
East Horizontal	791.4	218.6	371.1	32.8
West Horizontal	791.4	218.6	371.1	32.8
East Vertical	118.0	131.2	13.1	167.3
West Vertical	118.0	131.2	13.1	167.3
Estimated: Wiring Harness				
Cabling	2541.3	N/A	N/A	N/A
Total sum	49508.6	N/A	N/A	N/A

Table 10: Detailed weight and dimensions estimations

Components	Xcg (mm)	Ycg (mm)	Zcg (mm)
Avionics - basics			
Computer for flight control and processing X2 box	242.3	0.0	114.8
Sat comm	282.8	0.0	104.1
Microcontroller (python)X6 box	236.8	0.0	42.8
HUMS+Battery Management System	290.3	0.0	17.2
EASA anti-collision green light x2 box	600.7	0.0	negligible
Avionics - sensors			
Barometric Altimeter x2 box [inside Small avionics box (SAB)]	211.8	0.0	negligible
Engine Controller unit (ECU)	264.8	0.0	18.2
Thermal sensor X2 (engine mounted)	1002.4	0.0	16.39
Rpm sensor X5 (motor mounted)	600.7	0.0	negligible
Inertia measuring unit (IMU) x2 box (inside SAB)	211.8	0.0	negligible
Pitot airspeed sensor box (wing mounted)	600.7	0.0	negligible
Lidar Altimeter x2 box	146.8	0.0	73.25
AOA sensor (Pressure differential 4 pitot) x2 (wing mounted)	600.7	0.0	negligible
GNNS Receiver (multi-band) x2 box (inside SAB)	211.8	0.0	negligible
Thermal Camera	211.8	0.0	102.5
Avionics - actuators			
East Aileron servo1	658.9	0.7	-150.3
West Aileron servo2	658.9	-0.7	-150.3
East Rudder servo 1	1012.7	0.7	-127.0
West Rudder servo 2	1012.7	-0.7	-127.0
East Elevator servo 1	1021.5	0.5	-127.0
West Elevator servo 2	1021.5	-0.5	-127.0
East Payload mechanism servo 1	600.8	0.0	60.0
West Payload mechanism servo 2	600.8	0.0	60.0
U/C			
Landing Buoy front	405.0	0.0	142.2
Landing Buoy back	927.2	0.0	142.2
Payload			
Raft	600.8	0.0	60.0
Structural - fuselage			
Aluminium Structures + impact absorption foam floor	600.8	0.0	60.0
Structural - wing			
East	600.7	414.0	-150.3
West	600.7	-414.0	-150.3
Propulsion			
2 stroke IC Engine for back propeller (T-shaped)	1002.4	0.0	16.4
ESC VTOL 1	600.7	870.0	-150.3
Motors (brushless) 1	280.7	870.0	-123.0
VTOL prop 1	280.7	870.0	-167.7
ESC VTOL 2	600.7	-870.0	-150.3
Motors (brushless) 2	280.7	-870.0	-123.0
VTOL prop 2	280.7	-870.0	-167.7
ESC VTOL 3	600.7	870.0	-150.3
Motors (brushless) 3	920.7	870.0	-123.0
VTOL prop 3	920.7	870.0	-167.7
ESC VTOL 4	600.7	-870.0	-150.3
Motors (brushless) 4	920.7	-870.0	-123.0
VTOL prop 4	920.7	-870.0	-167.7
Pusher prop	1470.4	0.0	0.0
Battery Pack	340.3	0.0	78.0
APU	340.3	0.0	-54.0
Electric engine starter mounted aft of ICE	1112.9	0.0	0.0
Clutch	1146.9	0.0	0.0
Shaft	1309.9	0.0	0.0
Throttle Linear actuator	1002.4	0.0	16.4
Fuel+trapped Mount	844.1	0.0	60.0
Structural - Pylons			
East-Front	440.7	870.0	-132.2
West-Front	440.7	-870.0	-132.2
East-back	760.7	870.0	-132.2
West-back	760.7	-870.0	-132.2
Structural - H-tailplane			
East Horizontal	1220.2	60.9	-127.0
West Horizontal	1220.2	-60.9	-127.0
East Vertical	1225.5	148.8	-127.0
West Vertical	1225.5	-148.8	-127.0
Estimated: Wiring Harness			
Cabling	600.8	0.0	142.2
Total sum	N/A	N/A	N/A

Table 11: Components x,y,z relative position according to the final aircraft layout

Components	Moment X (kgmm)	Moment Y (kgmm)	Moment Z (kgmm)
Avionics - basics			
Computer for flight control and processing X2 box	91683	0	43421
Sat comm	120695	0	44430
Microcontroller (python)X6 box	39070	0	7062
HUMS+Battery Management System	95796	0	5676
EASA anti-collision green light x2 box	negligible	negligible	negligible
Avionics - sensors			
Barometric Altimeter x2 box [inside Small avionics box (SAB)]	negligible	negligible	negligible
Engine Controller unit (ECU)	24757.9	0.0	1701.7
Thermal sensor X2 (engine mounted)	negligible	negligible	negligible
Rpm sensor X5 (motor mounted)	negligible	negligible	negligible
Inertia measuring unit (IMU) x2 box (inside SAB)	negligible	negligible	negligible
Pitot airspeed sensor box (wing mounted)	negligible	negligible	negligible
Lidar Altimeter x2 box	11302.8	0.0	5640.3
AOA sensor (Pressure differential 4 pitot) x2 (wing mounted)	132154.6871	negligible	negligible
GNSS Receiver (multi-band) x2 box (inside SAB)	negligible	negligible	negligible
Thermal Camera	104836.05	0	50737.5
Avionics - actuators			
East Aileron servo1	44209	47	-10083
West Aileron servo2	44209	-47	-10083
East Rudder servo 1	37874	25	-4751
West Rudder servo 2	37874	-25	-4751
East Elevator servo 1	38204	17	-4751
West Elevator servo 2	38204	-17	-4751
East Payload mechanism servo 1	22470	0	2244
West Payload mechanism servo 2	22470	0	2244
U/C			
Landing Buoy front	222745	0	78210
Landing Buoy back	509955	0	78210
Payload			
Raft	5207648	0	520080
Structural - fuselage			
Aluminium Structures + impact absorption foam floor	2246955	0	224400
Structural - wing			
East	2246630	1548360	-561988
West	2246630	-1548360	-561988
Propulsion			
2 stroke IC Engine for back propeller (T-shaped)	3482303	0	56939
ESC VTOL 1	51540	74646	-12893
Motors (brushless) 1	194527	602910	-85225
VTOL prop 1	62055	192331	-37080
ESC VTOL 2	51540	-74646	-12893
Motors (brushless) 2	194527	-602910	-85225
VTOL prop 2	62055	-192331	-37080
ESC VTOL 3	51540	74646	-12893
Motors (brushless) 3	638047	602910	-85225
VTOL prop 3	203540	192331	-37080
ESC VTOL 4	51540	-74646	-12893
Motors (brushless) 4	638047	-602910	-85225
VTOL prop 4	203540	-192331	-37080
Pusher prop	85871	0	0
Battery Pack	2058755	0	471900
APU	262023	0	-41580
Electric engine starter mounted aft of ICE	865828	0	0
Clutch	580326	0	0
Shaft	82523	0	0
Throttle Linear actuator	88210	0	1442
Fuel+trapped Mount	4261810	0	302940
Structural - Pylons			
East-Front	78710	155382	-23616
West-Front	78710	-155382	-23616
East-back	135862	155382	-23616
West-back	135862	-155382	-23616
Structural - H-tailplane			
East Horizontal	965638	48163	-100528
West Horizontal	965638	-48163	-100528
East Vertical	144595	17561	-14988
West Vertical	144595	-17561	-14988
Estimated: Wiring Harness			
Cabling	1526771	0	361369
Total sum	30406128	0	-173730

Table 12: Detail component moment contribution estimation for calculating total cg

7.9 Detailed weight, cg and inertia estimation breakdown

Components	Global Ixx (g * mm ²)	Global Iyy (g * mm ²)	Global Ixy (g * mm ²)	Global Ixz (g * mm ²)	Global Izz (g * mm ²)
Avionics - basics					
Computer for flight control and processing X2 box	6579020	27484937	0	10520571	23582214
Sat comm	7896105	39289898	0	12564326	37046589
Microcontroller (python)X6 box	923825	9598431	0	1672211	9839331
HUMS+Battery Management System	536087	28045558	0	1647686	28174971
EASA anti-collision green light x2 box	negligible	negligible	negligible	negligible	negligible
Avionics - sensors					
Barometric Altimeter x2 box [inside Small avionics box (SAB)]	negligible	negligible	negligible	negligible	negligible
Engine Controller unit (ECU)	88014	6616651	0	450593	6586623
Thermal sensor X2 (engine mounted)	negligible	negligible	negligible	negligible	negligible
Rpm sensor X5 (motor mounted)	negligible	negligible	negligible	negligible	negligible
Inertia measuring unit (IMU) x2 box (inside SAB)	negligible	negligible	negligible	negligible	negligible
Pitot airspeed sensor box (wing mounted)	negligible	negligible	negligible	negligible	negligible
Lidar Altimeter x2 box	497799	2098549	0	827932	1729084
AOA sensor (Pressure differential 4 pitot) x2 (wing mounted)	negligible	negligible	negligible	negligible	negligible
GNSS Receiver (multi-band) x2 box (inside SAB)	negligible	to be designed	negligible	to be designed	negligible
Thermal Camera	6245250	27403821	0	10745695	23114852
Avionics - actuators					
East Aileron servo1	1524832	30653836	30946	-6643071	29144050
West Aileron servo2	1524832	30653836	-30946	-6643071	29144050
East Rudder servo 1	607214	38963872	24997	-4810974	38358287
West Rudder servo 2	607214	38963872	-24997	-4810974	38358287
East Elevator servo 1	607205	39632743	17650	-4852876	39031429
West Elevator servo 2	607205	39632743	-17650	-4852876	39031429
East Payload mechanism servo 1	138433	13637854	0	1348173	13505411
West Payload mechanism servo 2	138433	13637854	0	1348173	13505411
U/C					
Landing Buoy front	12715457	101679281	0	31674268	91803290
Landing Buoy back	12715457	484294699	0	72515530	474418708
Payload					
Raft	116729067	3159907474	0	312458863	3276879244
Structural - fuselage					
Aluminium Structures + impact absorption foam floor	479505400	2123149221	0	134817276	2109685221
Structural - wing					
East	1103221128	1472319920	930104688	-337587916	2405374205
West	1103221128	1472319920	-930104688	-337587916	2405374205
Propulsion					
2 stroke IC Engine for back propeller (T-shaped)	17308625	3504481748	0	57074944	3509279144
ESC VTOL 1	66889829	32946533	44840085	-7744664	95960321
Motors (brushless) 1	535498239	65570950	169238720	-23922813	579824472
VTOL prop 1	175234767	25325928	53987884	-10408444	188062976
ESC VTOL 2	66889829	32946533	-44840085	-7744664	95960321
Motors (brushless) 2	535498239	65570950	-169238720	-23922813	579824472
VTOL prop 2	175234767	25325928	-53987884	-10408444	188062976
ESC VTOL 3	66889829	32946533	44840085	-7744664	95960321
Motors (brushless) 3	535498239	598418649	555101120	-78466562	1112672171
VTOL prop 3	175234767	195306650	177079660	-34139581	358043699
ESC VTOL 4	66889829	32946533	-44840085	-7744664	95960321
Motors (brushless) 4	535498239	598418649	-555101120	-78466562	1112672171
VTOL prop 4	175234767	195306650	-177079660	-34139581	358043699
Pusher prop	876000	126705346	0	0	126705346
Battery Pack	78662912	753071940	0	160582851	730998318
APU	3738671	91768177	0	-14149258	90840520
Electric engine starter mounted aft of ICE	2107083	964299545	0	0	965109961
Clutch	1385428	666309110	0	0	666309110
Shaft	38588	108566145	0	0	108566145
Throttle Linear actuator	23640	88444782	0	1445767	88421143
Fuel+trapped Mount	186217218	3706724821	0	255708625	3688548421
Structural - Pylons					
East-Front	138390901	39377263	68477333	-10407707	171436859
West-Front	138390901	39377263	-68477333	-10407707	171436859
East-back	138390901	108040033	118199573	-17964872	240099629
West-back	138390901	108040033	-118199573	-17964872	240099629
Structural - H-tailplane					
East Horizontal	24851423	1194230372	58766810	-122660193	1193400901
West Horizontal	24851423	1194230372	-58766810	-122660193	1193400901
East Vertical	4794515	179545740	21521111	-18367140	179982206
West Vertical	4794515	179545740	-21521111	-18367140	179982206
Estimated: Wiring Harness					
Cabling	51386655	968655286	0	217106808	917268632
Total sum	6870334091	24153773881	0	-318188726	29515769894

Table 13: Detailed components self moment of inertia

7.10 Detailed weight, cg and inertia estimation breakdown

Components	Self Ixx (g * mm ²)	Self Iyy (g * mm ²)	Self Ixy (g * mm ²)	Self Ixz (g * mm ²)	Self Izz (g * mm ²)
Avionics - basics					
Computer for flight control and processing X2 box	1596414	288569	0	0	1368452
Sat comm	3270954	533473	0	0	2915315
Microcontroller (python)X6 box	621572	44709	0	0	587863
HUMS+Battery Management System	438460	139398	0	0	366438
EASA anti-collision green light x2 box	negligible	negligible	negligible	negligible	negligible
Avionics - sensors					
Barometric Altimeter x2 box [inside Small avionics box (SAB)]	negligible	negligible	negligible	negligible	negligible
Engine Controller unit (ECU)	57043	30045	0	0	30987
Thermal sensor X2 (engine mounted)	negligible	negligible	negligible	negligible	negligible
Rpm sensor X5 (motor mounted)	negligible	negligible	negligible	negligible	negligible
Inertia measuring unit (IMU) x2 box (inside SAB)	negligible	negligible	negligible	negligible	negligible
Pitot airspeed sensor box (wing mounted)	negligible	negligible	negligible	negligible	negligible
Lidar Altimeter x2 box	84650	26259	0	0	69942
AOA sensor (Pressure differential 4 pitot) x2 (wing mounted)	negligible	negligible	negligible	negligible	negligible
GNSS Receiver (multi-band) x2 box (inside SAB)	negligible	to be designed	negligible	to be designed	negligible
Thermal Camera	1044656	0	0	0	911625
Avionics - actuators					
East Aileron servo1	9727	11183	0	0	16437
West Aileron servo2	9727	11183	0	0	16437
East Rudder servo 1	3735	5933	0	0	3793
West Rudder servo 2	3735	5933	0	0	3793
East Elevator servo 1	3735	3793	0	0	5933
West Elevator servo 2	3735	3793	0	0	5933
East Payload mechanism servo 1	3793	3735	0	0	5933
West Payload mechanism servo 2	3793	3735	0	0	5933
U/C					
Landing Buoy front	1593995	348524	0	0	1593995
Landing Buoy back	1593995	348524	0	0	1593995
Payload					
Raft	85524267	0	0	0	148176571
Structural - fuselage					
Aluminium Structures + impact absorption foam floor	466041400	759737367	0	0	759737367
Structural - wing					
East	377753442	38315808	0	0	414795699
West	377753442	38315808	0	0	414795699
Propulsion					
2 stroke IC Engine for back propeller (T-shaped)	16375397	12922956	0	1234347	18653580
ESC VTOL 1	10503	48792	0	0	57865
Motors (brushless) 1	485683	485683	0	0	688362
VTOL prop 1	1687501	1687501	0	0	3316050
ESC VTOL 2	10503	48792	0	0	57865
Motors (brushless) 2	485683	485683	0	0	688362
VTOL prop 2	1687501	1687501	0	0	3316050
ESC VTOL 3	10503	48792	0	0	57865
Motors (brushless) 3	485683	485683	0	0	688362
VTOL prop 3	1687501	1687501	0	0	3316050
ESC VTOL 4	10503	48792	0	0	57865
Motors (brushless) 4	485683	485683	0	0	688362
VTOL prop 4	1687501	1687501	0	0	3316050
Pusher prop	876000	441815	0	0	441815
Battery Pack	41854712	15690171	0	0	30424749
APU	1493351	358948	0	0	1676611
Electric engine starter mounted aft of ICE	2107083	727754	0	0	1538171
Clutch	1385428	738634	0	0	738634
Shaft	38588	470001	0	0	470001
Throttle Linear actuator	0	0	0	0	0
Fuel+trapped Mount	168040818	91196872	0	0	91196872
Structural - Pylons					
East-Front	85817	1566962	0	0	1566962
West-Front	85817	1566962	0	0	1566962
East-back	85817	1566962	0	0	1566962
West-back	85817	1566962	0	0	1566962
Structural - H-tailplane					
East Horizontal	9150801	3223375	0	0	12232316
West Horizontal	9150801	3223375	0	0	12232316
East Vertical	276899	444487	0	0	170973
West Vertical	276899	444487	0	0	170973
Estimated: Wiring Harness					
Cabling	0	0	0	0	0
Total sum	1577521066	983216395	0	1234347	1939472103

Table 14: Detail component global moment of inertia using parallel axis theorem, combined for total MoI

7.11 Estimation Fidelity

Status	Description	WM Score
Allocation	An allocation is provided for a system, often without detailed analysis or substantiation. The allocation can take the form of a fixed weight or a simple mass fraction.	0
Parametric estimate	Parametric weight estimating relationships are used to estimate the weight of the component.	0.25
Detailed estimate	Analytically derived weight estimate based on detailed methods capturing the specific attributes of the component.	0.5
Vendor specification data	Weights data provided by the vendor based on their weights measurements.	0.5
Vendor contractual weight	Weight at which the vendor is contractually obligated to deliver the component.	0.75
Calculated	The weight is based on calculating the weight from a CAD model. In the case of structures, the structural geometry is defined as well as all skin thicknesses and material systems.	0.75
Measured	Component is weighed in the final configuration.	1.0

Figure 18: Weight Maturity Events and Associated Scores [13]

7.12 Matlab Script for Pusher Rotor spool up time Analysis

```
1
2 %%Pusher code
3 %only difference is that VTOL has V_inf lined up with omega
4 % unless travelling at unreasonable speeds, should V_inf have minimal
5 % effect
6 % model propeller drag for every rpm/speed, mainly for vtol since
7 % there is not much emphasis related to transition to spool up
8 % but to slow down
9
10 % NetTorque/Mass moment of inertia=Angular acceleration
11
12
13 load('PusherData.mat')
14 load('Pusher_distribution2.mat')
15 twist=pusher_struct.twist;
16 power_matrix_save=power_matrix;
17
18 power_matrix=power_matrix_save(:, :, 21);
19
20
21 [~, CL_max, LD_max] = Characteristics("E216Data.txt");
22 E216.CL_max = CL_max;
23 E216.LD_max = LD_max;
24
25 position=pusher_struct.positions;
26 chord=pusher_struct.chord;
27 twist=pusher_struct.twist;
28 last_circ=pusher_struct.last_circ;
29
30 %*****chop circular part
31 position=position(last_circ+1:end);
32 chord=chord(last_circ+1:end);
33
34 [af_upper, af_lower, t_max]=myAerofoil('E216Coordinates.txt');
35 af=[af_upper, af_lower];
36
37 plot(af(1,:), af(2,:));
38 hold on
39 xlim([0, 1])
40 ylim([-0.5 0.5])
41 hold off
42
43
44 airfoil_name = "e216";
45 t_c_max = 0.104; %REMEMBER to input thickness to chord ratio.
46 %Re.range = 50000:10000:410000;
47 Re.range = [50000, 100000, 200000, 500000, 1000000];
48
49 Re_min = Re.range(1);
50 Re_max = Re.range(length(Re.range));
51
52
53 rho = 1.225;
54 mu = 1.81*10^(-5);
55
56 interpolation_method = 'smoothingspline';
57
```



```

58 Re_plot_index = 2;
59
60 alpha_des_range = zeros(size(Re_range));
61 Cl_des_range = zeros(size(Re_range));
62 Cd_des_range = zeros(size(Re_range));
63 LD_max_range = zeros(size(Re_range));
64 alpha_mesh=[-80:0.25:80];
65 CL_mesh=zeros(length(Re_range),length(alpha_mesh));
66 CD_mesh=zeros(length(Re_range),length(alpha_mesh));
67 for i = 1:length(Re_range)
68     Re = Re_range(i);
69     % Use this for airfoiltools:
70     airfoil_aero_data = importdata("CSV/" + "xf-" + airfoil_name + "-il-" +
        num2str(Re) + ".csv",'',11);
71     Cl_alpha_fit = fit(airfoil_aero_data.data(:,1), airfoil_aero_data.data(:,2),
        interpolation_method);
72     Cd_alpha_fit = fit(airfoil_aero_data.data(:,1), airfoil_aero_data.data(:,3),
        interpolation_method);
73
74     Clm_alpha_fit = fit(airfoil_aero_data.data(:,1), airfoil_aero_data.data
        (:,5), interpolation_method);
75     LD_alpha = airfoil_aero_data.data(:,2) ./ airfoil_aero_data.data(:,3);
76     LD_max = max(LD_alpha);
77     alpha_des_index = find(LD_alpha == LD_max);
78     alpha_des = airfoil_aero_data.data(alpha_des_index,1);
79     Cl_des = airfoil_aero_data.data(alpha_des_index,2);
80     Cd_des = airfoil_aero_data.data(alpha_des_index,3);
81     alpha_data = airfoil_aero_data.data(:,1);
82     Cl_data = airfoil_aero_data.data(:,2);
83     Cd_data = airfoil_aero_data.data(:,3);
84
85     alpha_min_data = min(airfoil_aero_data.data(:,1));
86     alpha_max_data = max(airfoil_aero_data.data(:,1));
87
88     alpha_estimation_post = linspace(alpha_max_data+0.001, 90, 100);
89     alpha_estimation_pre = linspace(-90, alpha_min_data-0.001, 100);
90
91     Cl2max = 1.19*(1-t_c_max^2);
92     Cl_estimation_post = -0.032*(alpha_estimation_post-92)-(1.632-Cl2max)*...
93         ((92-alpha_estimation_post)/51).^(1+Cl2max/(1.632-Cl2max));
94
95     Cd1max = max(airfoil_aero_data.data(:,3));
96     Cd2max = 2.3*exp(-(0.65*t_c_max)^0.9);
97     Cd_estimation_post = Cd1max + (Cd2max-Cd1max)*...
98         sind((alpha_estimation_post-alpha_max_data)/(90-alpha_max_data)*90);
99
100     alpha_extended = [airfoil_aero_data.data(:,1)', alpha_estimation_post];
101     Cl_extended = [airfoil_aero_data.data(:,2)', Cl_estimation_post];
102     Cd_extended = [airfoil_aero_data.data(:,3)', Cd_estimation_post];
103     Cl_alpha_fit = fit(alpha_extended', Cl_extended', interpolation_method);
104     Cd_alpha_fit = fit(alpha_extended', Cd_extended', interpolation_method);
105
106     alpha_Cl_0 = fzero(Cl_alpha_fit, 0); %Check it's greater or equal to 0
107
108     Cl_estimation_pre = -Cl_alpha_fit(alpha_Cl_0-alpha_estimation_pre);
109     Cd_estimation_pre = Cd_alpha_fit(alpha_Cl_0-alpha_estimation_pre);
110     alpha_extended = [alpha_estimation_pre, alpha_extended];
111     Cl_extended = [Cl_estimation_pre', Cl_extended];
112     Cd_extended = [Cd_estimation_pre', Cd_extended];

```

```

113     Cl_alpha_fit = fit(alpha_extended', Cl_extended', interpolation_method);
114     Cd_alpha_fit = fit(alpha_extended', Cd_extended', interpolation_method);
115     Cl_alpha_fit_range{i} = fit(alpha_extended', Cl_extended',
        interpolation_method);
116     Cd_alpha_fit_range{i} = fit(alpha_extended', Cd_extended',
        interpolation_method);
117     LD_max_range(i) = LD_max;
118     alpha_des_range(i) = alpha_des;
119     Cl_des_range(i) = Cl_des;
120     Cd_des_range(i) = Cd_des;
121
122     CL_mesh(i,:) = Cl_alpha_fit(alpha_mesh);
123     CD_mesh(i,:) = Cd_alpha_fit(alpha_mesh);
124 end
125
126 Re_mesh = [30000:10000:1000000];
127 Re_CL_mesh = zeros(length(Re_mesh), size(CL_mesh, 2));
128 Re_CD_mesh = zeros(length(Re_mesh), size(CD_mesh, 2));
129 Re_CL_mesh2 = zeros(length(Re_mesh), size(CL_mesh, 2));
130 Re_CD_mesh2 = zeros(length(Re_mesh), size(CD_mesh, 2));
131 %turbulent 200000 mostly
132
133 for i = 1:size(CL_mesh, 2)
134     Re_CL_fit = fit(Re_range', CL_mesh(:, i), interpolation_method);
135     Re_CD_fit = fit(Re_range', CD_mesh(:, i), interpolation_method);
136     %Re_CL_fit2 = fit(Re_range', CL_mesh(:, i), poly, 6);
137     %Re_CD_fit2 = fit(Re_range', CD_mesh(:, i), poly, 6);
138     Re_CL_mesh(:, i) = Re_CL_fit(Re_mesh);
139     Re_CD_mesh(:, i) = Re_CD_fit(Re_mesh);
140     %Re_CL_mesh2(:, i) = Re_CL_fit2(Re_mesh);
141     %Re_CD_mesh2(:, i) = Re_CD_fit2(Re_mesh);
142 end
143
144
145
146 Re_mesh_matrix = repmat(Re_mesh, size(Re_CL_mesh, 2), 1)';
147 alpha_mesh_matrix = repmat(alpha_mesh, size(Re_CL_mesh, 1), 1);
148 hold on
149 hold off
150 plot3(alpha_mesh_matrix, Re_mesh_matrix, Re_CD_mesh)
151 title('C_D vs Re and Alpha')
152 xlabel('Alpha (deg)')
153 ylabel('Re')
154 zlabel('C_D')
155 hold on
156 hold off
157
158 hold on
159 hold off
160 surf(alpha_mesh, Re_mesh, Re_CD_mesh)
161 title('C_D vs Re and Alpha')
162 xlabel('Alpha (deg)')
163 ylabel('Re')
164 zlabel('C_D')
165 hold on
166 hold off
167
168 hold on
169 hold off

```

```

170 plot3(alpha_mesh_matrix, Re_mesh_matrix, Re_CL_mesh)
171 title('CL vs Re and Alpha')
172 xlabel('Alpha (deg)')
173 ylabel('Re')
174 zlabel('CL')
175 hold on
176 hold off
177
178 %problem is its not desgined to do low spd
179 % first interpolate Re, then find V and for each vinf of each section
180 % then use V and Vinf to calc alpha(with twist) and Re, then use Re and alpha
181 % to calculate Cd cl using interpolation, then use position at section to calc
182 % sectional drag with chord at position.
183 % sum along span all sections of blade using trapezoidal
184 % then lift drag modelled so aero torque, motor torque modelled,
185 % inertia from CAD then acceleration modelled
186
187
188
189 rad_per_s_range=rpm_range*0.10472;
190 rad_per_s_matrix= repmat(rad_per_s_range, length(V_inf_range), 1);
191 V_inf_matrix=transpose(repmat(V_inf_range, length(rpm_range), 1));
192 CD_matrix=zeros(100,46);
193 CL_matrix=zeros(100,46);
194 V_c_matrix=zeros(100,46);
195 Mach_matrix=zeros(100,46);
196 V_c_matrix=rad_per_s_matrix.*0.3;
197 V_total_matrix=sqrt(V_c_matrix.^2+V_inf_matrix.^2);
198 Mach_matrix=V_total_matrix/339.75; % at 400ft
199 % shows max mach of 0.56 %checked smaller than 0.56 at tip
200
201
202 %3-D Vtheta, Vinf, position, each entry corresponds to alpha and Re so CD
203 %each plane of a Vtheta and Vinf describes all regimes for one chord
204 blade_regime_3d=zeros(length(V_inf_range), length(rad_per_s_range), length(
    position));
205 V_total_3d=zeros(length(V_inf_range), length(rad_per_s_range), length(position));
206 alpha_total_3d=zeros(length(V_inf_range), length(rad_per_s_range), length(position)
    ));
207 Re_total_3d=zeros(length(V_inf_range), length(rad_per_s_range), length(position));
208 CL_total_3d=zeros(length(V_inf_range), length(rad_per_s_range), length(position));
209 CD_total_3d=zeros(length(V_inf_range), length(rad_per_s_range), length(position));
210 Torque_tangential=zeros(length(V_inf_range), length(rad_per_s_range), length(
    position));
211 Total_Torque_tangential=zeros(length(V_inf_range), length(rad_per_s_range));
212
213
214 rho=1.225; % ~AMSL
215 mu=1.81*10^-5; % 15 deg
216
217
218 a_dash=0;
219 a_normal=0;
220 for i=1:size(blade_regime_3d, 1)
221     for j=1:size(blade_regime_3d, 2)
222         for k=1:size(blade_regime_3d, 3)
223             % make Vtotal, Alpha and Re 3-D matrix
224             axial_ind=V_inf_range(i)/(2)+sqrt((V_inf_range(i)^2/4
                +51.5*9.81/4/2/1.225/(pi*0.3^2)));

```

```

225     V_circumferencial=rad_per_s_range(j)*position(k);
226     V_total_3d(i,j,k)=sqrt((V_inf_range(i)*(1+axial_ind))^2+(
        rad_per_s_range(j)*position(k))^2);
227 %for a_dash=0:0.05:1
228     % for a_normal=0:0.5:1
229     alpha_total_3d(i,j,k)=-atan(V_inf_range(i)*(1+axial_ind)/
        V_circumferencial)+twist(k);
230     Re_total_3d(i,j,k)=rho/mu*V_total_3d(i,j,k)*chord(k);
231
232 %now CL and CD interpolation
233     [~,ReIndex] = min(abs(Re_mesh-Re_total_3d(i,j,k)));
234     [~,alphaIndex] = min(abs(alpha_mesh-alpha_total_3d(i,j,k)));
235     CL_total_3d(i,j,k)=Re.CL_mesh(ReIndex,alphaIndex);
236     CD_total_3d(i,j,k)=Re.CD_mesh(ReIndex,alphaIndex);
237
238 % validate guess of a and a_dash
239 %a_normal=chord(k)/(8*pi*position(k)/2/CL_total_3d(i,j,k)-chord(k));
240 %a_dash=2*(CL_total_3d(i,j,k)*sin(alpha_total_3d(i,j,k))+CD_total_3d
        (i,j,k)*cos(alpha_total_3d(i,j,k)))*V_inf_range(i)*(1+a_normal)*
        chord(k)/8/pi/position(k)^3/(V_circumferencial/V_inf_range(i));
241
242
243 %then calculate sectional torque for each Vtheta V_inf
244 %combination across the span with different chord lengths
245
246     Torque_tangential(i,j,k)=(CL_total_3d(i,j,k)*1/2*rho*V_total_3d(i,j,
        k)^2*chord(k)*sin(alpha_total_3d(i,j,k))+CD_total_3d(i,j,k)*1/2*
        rho*V_total_3d(i,j,k)^2*chord(k)*cos(alpha_total_3d(i,j,k)))*2*(
        position(2)-position(1))*(position(k)-position(1));
247 %Aero Torque at each not even trapezoidal, just linear step
248 %addition, centre of action at centre of element
249
250 end
251
252 %sum for total along span for each Vtheta V_inf regime + viscous of
253 %typical motor
254     Total_Torque_tangential(i,j)=sum(Torque_tangential(i,j,:))+0.5*10^-3*
        rad_per_s_range(j);
255
256 end
257 end
258
259
260
261
262
263 rho=1.225;
264
265
266 %motor_torque_matrix=power_matrix./rad_per_s_matrix;
267 %maneuvre % no gearbox, rpm of prop=rpm of motor
268 %always at max power rpm for motor, but still P=T*omega
269
270 mass_blade=227.01; %g
271 centre_z=85.96; %mm
272 Izz_unshifted=857276.68; %gmm^2
273 Izz_shifted=Izz_unshifted+centre_z^2*mass_blade; %gmm^2
274 q=78000;
275

```

```

276 Rotor_Izz=617068.67; % gmm^2 fill in from CAD*****
277 Rotor_Izz=Rotor_Izz/1000^3
278
279 %motor_torque_matrix=power_matrix./rad_per_s_matrix;
280
281 %maneuvre % no gearbox, rpm of prop=rpm of motor
282 motor_torque_matrix=max(power_matrix)./rad_per_s_matrix; %with gearbox
283 motor_torque_matrix=flip(motor_torque_matrix,2);
284
285 motor_torque_matrix2=power_matrix./rad_per_s_matrix; %without gearbox
286 motor_torque_matrix2=flip(motor_torque_matrix2,2);
287
288 Net_torque_matrix=motor_torque_matrix-Total_Torque_tangential; % Nm
289 Net_torque_matrix2=motor_torque_matrix2-Total_Torque_tangential; % Nm%%
290
291 Angular_acceleration_matrix=Net_torque_matrix/Rotor_Izz; % rad/s^2
292 rpm_acceleration_matrix=Angular_acceleration_matrix*9.5493+q; %rpm/s^2
293
294 Angular_acceleration_matrix2=Net_torque_matrix2/Rotor_Izz; % rad/s^2
295 rpm_acceleration_matrix2=Angular_acceleration_matrix2*9.5493+q; %rpm/s^2
296
297 %0 torque, slow down
298 motor_torque_matrix3=0;
299 motor_torque_matrix3=flip(motor_torque_matrix3,2);
300
301 Net_torque_matrix3=motor_torque_matrix3-Total_Torque_tangential; % Nm
302
303 Angular_acceleration_matrix3=Net_torque_matrix3/Rotor_Izz; % rad/s^2
304 rpm_acceleration_matrix3=Angular_acceleration_matrix3*9.5493; %rpm/s^2
305
306 [xData, yData, zData] = prepareSurfaceData( rpm_range, V_inf_range,
      rpm_acceleration_matrix );
307
308 % Set up fitype and options.
309 ft = fitype( 'poly44' );
310
311 % Fit model to data.
312 [fitresult, gof] = fit( [xData, yData], zData, ft );
313 h = plot( fitresult );
314
315 %surf(rpm_range,V_inf_range,rpm_acceleration_matrix)
316 xlabel('Angular velocity (rev/min)');
317 ylabel('V-{\infty} (m/s)');
318 zlabel('Angular acceleration (rev/min^2)');
319 %title('Pusher propeller angular acceleration vs angular velocity and free-
      stream velocity');
320
321 surf(rpm_range,V_inf_range,rpm_acceleration_matrix2)
322 xlabel('rpm (rev/min)');
323 ylabel('V-{\inf} (m/s)');
324 zlabel('rpm acceleration (rev/min^2)');
325 title('VTOL:Rpm derivative variation with rpm and V-{\inf}');
326
327 surf(rpm_range,V_inf_range,rpm_acceleration_matrix3)
328 xlabel('rpm (rev/min)');
329 ylabel('V-{\inf} (m/s)');
330 zlabel('rpm acceleration (rev/min^2)');
331 title('VTOL:Rpm derivative variation with rpm and V-{\inf}');
332

```

```

333
334 plot(rpm_range, rpm_acceleration_matrix(1,:))
335 xlabel('rpm (rev/min)');
336 ylabel('rpm acceleration (rev/min^2)');
337 title('VTOL:Rpm derivative variation with rpm');
338
339 plot(rpm_range, rpm_acceleration_matrix2(1,:))
340 xlabel('rpm (rev/min)');
341 ylabel('rpm acceleration (rev/min^2)');
342 title('VTOL:Rpm derivative variation with rpm');
343
344 plot(rpm_range, rpm_acceleration_matrix3(1,:))
345 xlabel('rpm (rev/min)');
346 ylabel('rpm acceleration (rev/min^2)');
347 title('VTOL:Rpm derivative variation with rpm');
348
349 QQ1 = diff(rpm_range);% dv dv/dv*dt, integrate.
350 time=zeros(size(rpm_acceleration_matrix2,1),length(QQ1));
351 timedisc=zeros(size(rpm_acceleration_matrix2,1),length(QQ1));
352 for i=1:size(rpm_acceleration_matrix2,1)
353     time(i,:)=QQ1./(rpm_acceleration_matrix2(i,1:end-1))*60; %s
354     timedisc(i,:)=cumsum(time(i,:));
355 end
356
357 plot(timedisc(1,:), rpm_range(1:end-1))
358 xlabel('Time (s)');
359 ylabel('Angular velocity (rev/min)');
360 %title('Pusher propeller min to max rpm acceleration (level flight)');
361
362
363
364 step_response_time=timedisc((1:end-2),end);
365 step_response_time=transpose(step_response_time);
366
367 step_response_time2=zeros(length(step_response_time)-2,1)
368 function111=fit(V_inf_range(1:end-2),step_response_time','poly4')
369
370 step_response_time2=function111(linspace(0,V_inf_range(end-2),50));
371
372 plot(linspace(0,V_inf_range(end-2),50),step_response_time2)
373 ylabel('Time (s)');
374 xlabel('V_{\infty} (m/s)');
375 %title('Pusher propeller acceleration time vs free stream velocity (level flight)');
376 %makes sense, designed to have less 'drag' uring forward flight
377
378 pusher_for.time=timedisc(1,:)
379 pusher_for.rpm=rpm_range(1:end-1)
380 save('C:\Users\leewe\Downloads\josh\pusher_for.mat','pusher_for')
381
382 %%
383 %VTOL code
384 % NetTorque/Mass moment of inertia=Angular acceleration
385
386
387 load('VTOLData.mat')
388 load('VTOL_distribution.mat')
389
390

```

```

391
392 [~, CL_max, LD_max] = Characteristics("E216Data.txt");
393 E216.CL_max = CL_max;
394 E216.LD_max = LD_max;
395
396 position=VTOL_struct.positions;
397 chord=VTOL_struct.chord;
398 twist=VTOL_struct.twist;
399 last_circ=VTOL_struct.last_circ;
400
401 %*****chop circular part
402 position=position(last_circ+1:end);
403 chord=chord(last_circ+1:end);
404
405 [af_upper, af_lower, t_max]=myAerofoil('E216Coordinates.txt');
406 af=[af_upper, af_lower];
407
408 plot(af(1,:), af(2,:));
409 hold on
410 xlim([0,1])
411 ylim([-0.5 0.5])
412 hold off
413
414
415 airfoil_name = "e216";
416 t_c_max = 0.104; %REMEMBER to input thickness to chord ratio.
417 %Re_range = 50000:10000:410000;
418 Re_range = [50000, 100000, 200000, 500000, 1000000];
419
420 Re_min = Re_range(1);
421 Re_max = Re_range(length(Re_range));
422
423
424 rho = 1.225;
425 mu = 1.81*10^(-5);
426
427 interpolation_method = 'smoothingspline';
428
429 Re_plot_index = 2;
430
431 alpha_des_range = zeros(size(Re_range));
432 Cl_des_range = zeros(size(Re_range));
433 Cd_des_range = zeros(size(Re_range));
434 LD_max_range = zeros(size(Re_range));
435 alpha_mesh=[-80:0.25:80];
436 CL_mesh=zeros(length(Re_range),length(alpha_mesh));
437 CD_mesh=zeros(length(Re_range),length(alpha_mesh));
438 for i = 1:length(Re_range)
439     Re = Re_range(i);
440     % Use this for airfoiltools:
441     airfoil_aero_data = importdata("CSV/" + "xf-" + airfoil_name + "-il-" +
        num2str(Re) + ".csv", ',', 11);
442     Cl_alpha_fit = fit(airfoil_aero_data.data(:,1), airfoil_aero_data.data(:,2),
        interpolation_method);
443     Cd_alpha_fit = fit(airfoil_aero_data.data(:,1), airfoil_aero_data.data(:,3),
        interpolation_method);
444
445     Clm_alpha_fit = fit(airfoil_aero_data.data(:,1), airfoil_aero_data.data
       (:,5), interpolation_method);

```

```

446 LD_alpha = airfoil_aero_data.data(:,2)./airfoil_aero_data.data(:,3);
447 LD_max = max(LD_alpha);
448 alpha_des_index = find(LD_alpha == LD_max);
449 alpha_des = airfoil_aero_data.data(alpha_des_index,1);
450 Cl_des = airfoil_aero_data.data(alpha_des_index,2);
451 Cd_des = airfoil_aero_data.data(alpha_des_index,3);
452 alpha_data = airfoil_aero_data.data(:,1);
453 Cl_data = airfoil_aero_data.data(:,2);
454 Cd_data = airfoil_aero_data.data(:,3);
455
456 alpha_min_data = min(airfoil_aero_data.data(:,1));
457 alpha_max_data = max(airfoil_aero_data.data(:,1));
458
459 alpha_estimation_post = linspace(alpha_max_data+0.001, 90, 100);
460 alpha_estimation_pre = linspace(-90, alpha_min_data-0.001, 100);
461
462 Cl2max = 1.19*(1-t_c_max^2);
463 Cl_estimation_post = -0.032*(alpha_estimation_post-92)-(1.632-Cl2max)*...
464 ((92-alpha_estimation_post)/51).^(1+Cl2max/(1.632-Cl2max));
465
466 Cd1max = max(airfoil_aero_data.data(:,3));
467 Cd2max = 2.3*exp(-(0.65*t_c_max)^0.9);
468 Cd_estimation_post = Cd1max + (Cd2max-Cd1max)*...
469 sind((alpha_estimation_post-alpha_max_data)/(90-alpha_max_data)*90);
470
471 alpha_extended = [airfoil_aero_data.data(:,1)', alpha_estimation_post];
472 Cl_extended = [airfoil_aero_data.data(:,2)', Cl_estimation_post];
473 Cd_extended = [airfoil_aero_data.data(:,3)', Cd_estimation_post];
474 Cl_alpha_fit = fit(alpha_extended', Cl_extended', interpolation_method);
475 Cd_alpha_fit = fit(alpha_extended', Cd_extended', interpolation_method);
476
477 alpha_Cl_0 = fzero(Cl_alpha_fit, 0); %Check it's greater or equal to 0
478
479 Cl_estimation_pre = -Cl_alpha_fit(alpha_Cl_0-alpha_estimation_pre);
480 Cd_estimation_pre = Cd_alpha_fit(alpha_Cl_0-alpha_estimation_pre);
481 alpha_extended = [alpha_estimation_pre, alpha_extended];
482 Cl_extended = [Cl_estimation_pre, Cl_extended];
483 Cd_extended = [Cd_estimation_pre, Cd_extended];
484 Cl_alpha_fit = fit(alpha_extended', Cl_extended', interpolation_method);
485 Cd_alpha_fit = fit(alpha_extended', Cd_extended', interpolation_method);
486 Cl_alpha_fit_range{i} = fit(alpha_extended', Cl_extended',
487 interpolation_method);
488 Cd_alpha_fit_range{i} = fit(alpha_extended', Cd_extended',
489 interpolation_method);
490 LD_max_range(i) = LD_max;
491 alpha_des_range(i) = alpha_des;
492 Cl_des_range(i) = Cl_des;
493 Cd_des_range(i) = Cd_des;
494
495 CL_mesh(i,:) = Cl_alpha_fit(alpha_mesh);
496 CD_mesh(i,:) = Cd_alpha_fit(alpha_mesh);
497
498 end
499
500 Re_mesh = [30000:10000:1000000];
501 Re_CL_mesh = zeros(length(Re_mesh), size(CL_mesh, 2));
502 Re_CD_mesh = zeros(length(Re_mesh), size(CD_mesh, 2));
503 Re_CL_mesh2 = zeros(length(Re_mesh), size(CL_mesh, 2));
504 Re_CD_mesh2 = zeros(length(Re_mesh), size(CD_mesh, 2));
505 %turbulent 200000 mostly

```



```

503
504 for i=1:size(CL_mesh,2)
505     Re_CL_fit = fit(Re_range',CL_mesh(:,i), interpolation_method);
506     Re_CD_fit = fit(Re_range',CD_mesh(:,i), interpolation_method);
507     %Re_CL_fit2 = fit(Re_range',CL_mesh(:,i), poly,6);
508     %Re_CD_fit2 = fit(Re_range',CD_mesh(:,i), poly,6);
509     Re_CL_mesh(:,i)=Re_CL_fit(Re_mesh);
510     Re_CD_mesh(:,i)=Re_CD_fit(Re_mesh);
511     %Re_CL_mesh2(:,i)=Re_CL_fit2(Re_mesh);
512     %Re_CD_mesh2(:,i)=Re_CD_fit2(Re_mesh);
513 end
514
515
516 Re_mesh_matrix= repmat(Re_mesh, size(Re_CL_mesh,2),1)';
517 alpha_mesh_matrix= repmat(alpha_mesh, size(Re_CL_mesh,1),1);
518 hold on
519 hold off
520 plot3(alpha_mesh_matrix, Re_mesh_matrix, Re_CD_mesh)
521 title('CD vs Re and Alpha')
522 xlabel('Alpha (deg)')
523 ylabel('Re')
524 zlabel('CD')
525 hold on
526 hold off
527
528
529 hold on
530 hold off
531 surf(alpha_mesh, Re_mesh, Re_CD_mesh)
532 title('CD vs Re and Alpha')
533 xlabel('Alpha (deg)')
534 ylabel('Re')
535 zlabel('CD')
536 hold on
537 hold off
538
539 hold on
540 hold off
541 plot3(alpha_mesh_matrix, Re_mesh_matrix, Re_CL_mesh)
542 title('CL vs Re and Alpha')
543 xlabel('Alpha (deg)')
544 ylabel('Re')
545 zlabel('CL')
546 hold on
547 hold off
548
549 %problem is its not desgined to do low spd
550 % first interpolate Re, then find V and for each vinf of each section
551 % then use V and Vinf to calc alpha(with twist) and Re, then use Re and alpha
552 % to calculate Cd cl using interpolation, then use position at section to calc
553 % sectional drag with chord at position.
554 % sum along span all sections of blade using trapezoidal
555 % then lift drag modelled so aero torque, motor torque modelled,
556 % inertia from CAD then acceleration modelled
557
558
559
560 rad_per_s_range=rpm_range*0.10472;
561 rad_per_s_matrix= repmat(rad_per_s_range, length(V_inf_range),1);

```

```

562 V_inf_matrix=transpose(repmat(V_inf_range,length(rpm_range),1));
563 CD_matrix=zeros(100,46);
564 CL_matrix=zeros(100,46);
565 V_c_matrix=zeros(100,46);
566 Mach_matrix=zeros(100,46);
567 V_c_matrix=rad_per_s_matrix.*0.3;
568 V_total_matrix=sqrt(V_c_matrix.^2+V_inf_matrix.^2);
569 Mach_matrix=V_total_matrix/339.75; % at 400ft
570 % shows max mach of 0.56 %checked smaller than 0.56 at tip
571
572
573 %3-D Vtheta, Vinf, position, each entry corresponds to alpha and Re so CD
574 %each plane of a Vtheta and Vinf describes all regimes for one chord
575 blade_regime_3d=zeros(length(V_inf_range),length(rad_per_s_range),length(
    position));
576 V_total_3d=zeros(length(V_inf_range),length(rad_per_s_range),length(position));
577 alpha_total_3d=zeros(length(V_inf_range),length(rad_per_s_range),length(position)
    ));
578 Re_total_3d=zeros(length(V_inf_range),length(rad_per_s_range),length(position));
579 CL_total_3d=zeros(length(V_inf_range),length(rad_per_s_range),length(position));
580 CD_total_3d=zeros(length(V_inf_range),length(rad_per_s_range),length(position));
581 Torque_tangential=zeros(length(V_inf_range),length(rad_per_s_range),length(
    position));
582 Total_Torque_tangential=zeros(length(V_inf_range),length(rad_per_s_range));
583
584
585 rho=1.225; % ~AMSL
586 mu=1.81*10^-5; % 15 deg
587
588 for i=1:size(blade_regime_3d,1)
589     for j=1:size(blade_regime_3d,2)
590         for k=1:size(blade_regime_3d,3)
591             % make Vtotal, Alpha and Re 3-D matrix
592             %axial_ind=-V_inf_range(i)/(2)+sqrt((V_inf_range(i)^2/4
                +51.5*9.81/4/2/1.225/(pi*0.3^2)));
593             axial_ind=0;
594             V_circumferencial=rad_per_s_range(j)*position(k);
595             V_total_3d(i,j,k)=sqrt((V_inf_range(i)*(1+axial_ind))^2+(
                rad_per_s_range(j)*position(k))^2);
596             alpha_total_3d(i,j,k)=atan(V_inf_range(i)*(1+axial_ind)/
                V_circumferencial)+twist(k);
597             Re_total_3d(i,j,k)=rho/mu*V_total_3d(i,j,k)*chord(k);
598
599             %now CL and CD interpolation
600             [~,ReIndex] = min(abs(Re_mesh-Re_total_3d(i,j,k)));
601             [~,alphaIndex] = min(abs(alpha_mesh-alpha_total_3d(i,j,k)));
602             CL_total_3d(i,j,k)=Re_CL_mesh(ReIndex,alphaIndex);
603             CD_total_3d(i,j,k)=Re_CD_mesh(ReIndex,alphaIndex);
604
605             %then calculate sectional torque for each Vtheta V_inf
606             %combination across the span with different chord lengths
607
608             Torque_tangential(i,j,k)=(CL_total_3d(i,j,k)*1/2*rho*V_total_3d(i,j,
                k)^2*chord(k)*sin(alpha_total_3d(i,j,k))+CD_total_3d(i,j,k)*1/2*
                rho*V_total_3d(i,j,k)^2*chord(k)*cos(alpha_total_3d(i,j,k)))*2*(
                position(2)-position(1))*(position(k)-position(1));
609             %Aero Torque at each not even trapezoidal, just linear step
610             %addition, centre of action at centre of element
611

```

```

612         end
613
614         %sum for total along span for each Vtheta V_infregime
615         Total_Torque_tangential(i,j)=sum(Torque_tangential(i,j,:))+0.5*10^-3*
            rad_per_s_range(j);
616
617     end
618 end
619
620 rho=1.225;
621
622
623
624
625 %motor_torque_matrix=power_matrix./rad_per_s_matrix;
626 %maneuvre % no gearbox, rpm of prop=rpm of motor
627 %always at max power rpm for motor, but still P=T*omega
628
629 mass_blade=227.01; %g
630 centre_z=85.96; %mm
631 Izz_unshifted=857276.68; %gmm^2
632 Izz_shifted=Izz_unshifted+centre_z^2*mass_blade; %gmm^2
633 q=15000;
634
635 Rotor_Izz=2534583.28; %gmm^2 fill in from CAD*****
636 Rotor_Izz=Rotor_Izz/1000^3
637 %motor_torque_matrix=power_matrix./rad_per_s_matrix;
638
639 %maneuvre % no gearbox, rpm of prop=rpm of motor
640 motor_torque_matrix=max(power_matrix)./rad_per_s_matrix; %with gearbox
641 motor_torque_matrix=flip(motor_torque_matrix,2);
642
643 motor_torque_matrix2=power_matrix./rad_per_s_matrix; %without gearbox
644 motor_torque_matrix2=flip(motor_torque_matrix2,2);
645
646 Net_torque_matrix=motor_torque_matrix-Total_Torque_tangential; % Nm
647 Net_torque_matrix2=motor_torque_matrix2-Total_Torque_tangential; % Nm%%
648
649 Angular_acceleration_matrix=Net_torque_matrix/Rotor_Izz; % rad/s^2
650 rpm_acceleration_matrix=Angular_acceleration_matrix*9.54938+q; %rpm/s^2
651
652 Angular_acceleration_matrix2=Net_torque_matrix2/Rotor_Izz; % rad/s^2
653 rpm_acceleration_matrix2=Angular_acceleration_matrix2*9.5493+q; %rpm/s^2
654
655 %0 torque, slow down
656 motor_torque_matrix3=0;
657 motor_torque_matrix3=flip(motor_torque_matrix3,2);
658
659 Net_torque_matrix3=motor_torque_matrix3-Total_Torque_tangential; % Nm
660
661 Angular_acceleration_matrix3=Net_torque_matrix3/Rotor_Izz; % rad/s^2
662 rpm_acceleration_matrix3=Angular_acceleration_matrix3*9.5493; %rpm/s^2
663
664
665 surf(rpm_range,V_inf_range,rpm_acceleration_matrix)
666 xlabel('Angular velocity (rev/min)');
667 ylabel('V_{\infty} (m/s)');
668 zlabel('Angular acceleration (rev/min^2)');

```

```

669 %title('Pusher propeller angular acceleration vs angular velocity and free-
      stream velocity');
670
671 surf(rpm_range,V_inf_range,rpm_acceleration_matrix2)
672 xlabel('rpm (rev/min)');
673 ylabel('V-{\infty} (m/s)');
674 zlabel('rpm acceleration (rev/min^2)');
675 title('VTOL:Rpm derivative variation with rpm and V-{\infty}');
676
677 surf(rpm_range,V_inf_range,rpm_acceleration_matrix3)
678 xlabel('rpm (rev/min)');
679 ylabel('V-{inf} (m/s)');
680 zlabel('rpm acceleration (rev/min^2)');
681 title('VTOL:Rpm derivative variation with rpm and V-{inf}');
682
683
684 plot(rpm_range,rpm_acceleration_matrix(1,:))
685 xlabel('rpm (rev/min)');
686 ylabel('rpm acceleration (rev/min^2)');
687 title('VTOL:Rpm derivative variation with rpm');
688
689 plot(rpm_range,rpm_acceleration_matrix2(1,:))
690 xlabel('rpm (rev/min)');
691 ylabel('rpm acceleration (rev/min^2)');
692 title('VTOL:Rpm derivative variation with rpm');
693
694 plot(rpm_range,rpm_acceleration_matrix3(1,:))
695 xlabel('rpm (rev/min)');
696 ylabel('rpm acceleration (rev/min^2)');
697 title('VTOL:Rpm derivative variation with rpm');
698
699 QQ1 = diff(rpm_range);% dv dv/dv*dt, integrate.
700 time=zeros(size(rpm_acceleration_matrix2,1),length(QQ1));
701 timedisc=zeros(size(rpm_acceleration_matrix2,1),length(QQ1));
702 for i=1:size(rpm_acceleration_matrix2,1)
703     time(i,:)=QQ1./(rpm_acceleration_matrix2(i,1:end-1))*60; %s
704     timedisc(i,:)=cumsum(time(i,:));
705 end
706
707 plot(timedisc(51,:),rpm_range(1:end-1))
708 xlabel('Time (s)');
709 ylabel('Angular velocity (rev/min)');
710 %title('Pusher propeller min to max rpm acceleration (level flight)');
711
712 (38:60)
713
714 xx=30;
715 yy=95;
716
717 step_response_time=timedisc((1:end-2),end);
718 step_response_time=transpose(step_response_time);
719
720 step_response_time2=zeros(length(step_response_time)-2,1)
721 functionl11=fit(V_inf_range(xx:yy)',step_response_time(xx:yy)', 'poly4')
722
723 step_response_time2=functionl11(linspace(V_inf_range(xx),V_inf_range(yy),50));
724
725 plot(linspace(V_inf_range(xx),V_inf_range(yy),50),step_response_time2)
726 ylabel('Time (s)');

```

```

727 xlabel( 'V_{\infty} (m/s) ');
728 %%
729 %airfoil centroid code
730 af=[19.4      0
731 19.316968     0.011058
732 19.069424     0.042292
733 18.661636     0.089822
734 18.100394     0.14938
735 17.395592     0.218638
736 16.55887      0.295268
737 15.604972     0.37733
738 14.55         0.462496
739 13.411996     0.547662
740 12.210554     0.629918
741 10.966044     0.705772
742 9.7           0.771732
743 8.433956      0.824112
744 7.189446      0.859614
745 6.582032      0.869896
746 5.988004      0.874746
747 5.409884      0.873776
748 4.85          0.866404
749 4.310874      0.853018
750 3.795028      0.83323
751 3.304402      0.807234
752 2.84113       0.774836
753 2.407152      0.73623
754 2.004408      0.691416
755 1.634838      0.64117
756 1.299606      0.586462
757 1.000264      0.52768
758 0.738364      0.46463
759 0.514682      0.395566
760 0.330576      0.319324
761 0.186434      0.235516
762 0.083032      0.148798
763 0.020758      0.067706
764 0             0
765 0.020758      -0.067706
766 0.083032      -0.148798
767 0.186434      -0.235516
768 0.330576      -0.319324
769 0.514682      -0.395566
770 0.738364      -0.46463
771 1.000264      -0.52768
772 1.299606      -0.586462
773 1.634838      -0.64117
774 2.004408      -0.691416
775 2.407152      -0.73623
776 2.84113       -0.774836
777 3.304402      -0.807234
778 3.795028      -0.83323
779 4.310874      -0.853018
780 4.85          -0.866404
781 5.409884      -0.873776
782 5.988004      -0.874746
783 6.582032      -0.869896
784 7.189446      -0.859614
785 8.433956      -0.824112

```

```

786 9.7      -0.771732
787 10.966044      -0.705772
788 12.210554      -0.629918
789 13.411996      -0.547662
790 14.55      -0.462496
791 15.604972      -0.37733
792 16.55887      -0.295268
793 17.395592      -0.218638
794 18.100394      -0.14938
795 18.661636      -0.089822
796 19.069424      -0.042292
797 19.316968      -0.011058
798 19.4      0];
799
800 af=af/19.4*0.194
801
802 afx=af(:,1);
803 afy=af(:,2);
804
805 plot(afx,afy)
806 xlim([0 1])
807 ylim([-0.5 0.5])
808
809 %title('Pusher propeller acceleration time vs free stream velocity (level flight
      )');
810 %makes sense, designed to have less 'drag' uring forward flight
811
812
813
814
815 %title('VTOL propeller min to max rpm acceleration time vs free-stream velocity
      (upward flight)');
816 %makes sense, designed to have minimal drag when V_inf=0
817
818 % as jordi analysed, there is a change in the airfoil behaviour
819 % in performance, however, it is still a reasonable approximation to
820 % use a fixed reynolds number that is close to our flight regimes.
821 % because motor behaviour can also be unpredictable due
822 % to gusts loads on the aircraft.
823 % ranges from 50000 to 330000
824 % no double counting with matan
825
826
827
828 %Re 10^6
829 timedisc(51,:),rpm_range(1:end-1)
830 VTOL_up.time=timedisc(51,:)
831 VTOL_up.rpm=rpm_range(1:end-1)
832 save('C:\Users\leewe\Downloads\josh\vtol-up.mat','VTOL-up')
833
834
835
836 function [Curves, CL_max, LD_max] = Characteristics(filename)
837
838 Data = importdata(filename);
839
840 alpha = Data.data(:, 1) * pi / 180;
841 CL = Data.data(:, 2);

```

```

842 CD = Data.data(:, 3);
843
844 Curves.Lift_Curve = fit(alpha, CL, 'smoothing spline', 'smoothingparam', 0.9999);
845 Curves.Drag_Curve = fit(alpha, CD, 'smoothing spline', 'smoothingparam', 0.9999);
846
847 vals = min(alpha):0.01:max(alpha);
848 vals = vals';
849
850 CL_vals = Curves.Lift_Curve(vals);
851 CL_vals = double(CL_vals);
852 CL_max(1) = max(CL_vals);
853
854 CL_max(2) = sum(vals .* (CL_vals == CL_max(1)));
855
856 CD_vals = Curves.Drag_Curve(vals);
857 CD_vals = double(CD_vals);
858 LD = CL_vals ./ CD_vals;
859 LD_max(1) = max(LD);
860
861 LD_max(2) = sum(vals .* (LD == LD_max(1)));
862
863 end
864
865
866 function [cu, cl, t_max] = myAerofoil(filename)
867 % This function outputs the x and y coordinates for an inputted text file
868 % where column 1 are the x coordinates and column 2 are the y coordinates
869 % (the output from xfoil)
870
871 Data = importdata(filename); % imports file data
872
873 Data = Data.data; % extracts numerical data
874
875 n = size(Data);
876
877 n = n(1); % extracts the 'height' rather than the 'width'
878
879 % First number is 16. for some reason, which needs to be ignored
880 x = Data(1:n, 1);
881 y = Data(1:n, 2);
882
883 con = find(min(x) == x);
884
885 cu = [x(1:con)'; y(1:con)'];
886 cl = [x(con + 1:end)'; y(con + 1:end)'];
887
888 fu = fit(cu(1, :)', cu(2, :)', 'cubicinterp');
889 fl = fit(cl(1, :)', cl(2, :)', 'cubicinterp');
890
891 x = 0:0.01:1;
892
893 tu = fu(x);
894 tl = fl(x);
895
896 t = double(tu - tl);
897 t_max(1) = max(t);
898
899 t_max(2) = sum(x' .* (t == t_max(1)));
900

```

```

901 end
902
903 function [F] = Prandtl(B, R, r, phi)
904
905 F = 2 / pi * acos(exp(- B * (R - r) / (2 * r * sin(phi))));
906
907 end
908 function [ geom, iner, cpmo ] = polygeom( x, y )
909 %POLYGEOM Geometry of a planar polygon
910 %
911 % POLYGEOM( X, Y ) returns area, X centroid,
912 % Y centroid and perimeter for the planar polygon
913 % specified by vertices in vectors X and Y.
914 %
915 % [ GEOM, INER, CPMO ] = POLYGEOM( X, Y ) returns
916 % area, centroid, perimeter and area moments of
917 % inertia for the polygon.
918 % GEOM = [ area X_cen Y_cen perimeter ]
919 % INER = [ Ixx Iyy Ixy Iuu Ivv Iuv ]
920 % u,v are centroidal axes parallel to x,y axes.
921 % CPMO = [ I1 ang1 I2 ang2 J ]
922 % I1,I2 are centroidal principal moments about axes
923 % at angles ang1,ang2.
924 % ang1 and ang2 are in radians.
925 % J is centroidal polar moment. J = I1 + I2 = Iuu + Ivv
926
927 % H.J. Sommer III - 16.12.09 - tested under MATLAB v9.0
928 %
929 % sample data
930 % x = [ 2.000 0.500 4.830 6.330 ]';
931 % y = [ 4.000 6.598 9.098 6.500 ]';
932 % 3x5 test rectangle with long axis at 30 degrees
933 % area=15, x_cen=3.415, y_cen=6.549, perimeter=16
934 % Ixx=659.561, Iyy=201.173, Ixy=344.117
935 % Iuu=16.249, Ivv=26.247, Iuv=8.660
936 % I1=11.249, ang1=30deg, I2=31.247, ang2=120deg, J=42.496
937 %
938 % H.J. Sommer III, Ph.D., Professor of Mechanical Engineering, 337 Leonhard Bldg
939 % The Pennsylvania State University, University Park, PA 16802
940 % (814)863-8997 FAX (814)865-9693 hjs1-at-psu.edu www.mne.psu.edu/sommer/
941
942 % begin function POLYGEOM
943
944 % check if inputs are same size
945 if ~isequal( size(x), size(y) ),
946     error( 'X and Y must be the same size' );
947 end
948
949 % temporarily shift data to mean of vertices for improved accuracy
950 xm = mean(x);
951 ym = mean(y);
952 x = x - xm;
953 y = y - ym;
954
955 % summations for CCW boundary
956 xp = x( [2:end 1] );
957 yp = y( [2:end 1] );
958 a = x.*yp - xp.*y;
959

```



```

960 A = sum( a ) /2;
961 xc = sum( (x+xp).*a ) /6/A;
962 yc = sum( (y+yp).*a ) /6/A;
963 Ixx = sum( (y.*y +y.*yp + yp.*yp).*a ) /12;
964 Iyy = sum( (x.*x +x.*xp + xp.*xp).*a ) /12;
965 Ixy = sum( (x.*yp +2*x.*y +2*xp.*yp + xp.*y).*a ) /24;
966
967 dx = xp - x;
968 dy = yp - y;
969 P = sum( sqrt( dx.*dx +dy.*dy ) );
970
971 % check for CCW versus CW boundary
972 if A < 0,
973     A = -A;
974     Ixx = -Ixx;
975     Iyy = -Iyy;
976     Ixy = -Ixy;
977 end
978
979 % centroidal moments
980 Iuu = Ixx - A*yc*yc;
981 Ivv = Iyy - A*xc*xc;
982 Iuv = Ixy - A*xc*yc;
983 J = Iuu + Ivv;
984
985 % replace mean of vertices
986 x_cen = xc + xm;
987 y_cen = yc + ym;
988 Ixx = Iuu + A*y_cen*y_cen;
989 Iyy = Ivv + A*x_cen*x_cen;
990 Ixy = Iuv + A*x_cen*y_cen;
991
992 % principal moments and orientation
993 I = [ Iuu  -Iuv ;
994       -Iuv  Ivv ];
995 [ eig_vec , eig_val ] = eig(I);
996 I1 = eig_val(1,1);
997 I2 = eig_val(2,2);
998 ang1 = atan2( eig_vec(2,1), eig_vec(1,1) );
999 ang2 = atan2( eig_vec(2,2), eig_vec(1,2) );
1000
1001 % return values
1002 geom = [ A  x_cen  y_cen  P ];
1003 iner = [ Ixx  Iyy  Ixy  Iuu  Ivv  Iuv ];
1004 cpmo = [ I1  ang1  I2  ang2  J ];
1005
1006 % bottom of polygeom
1007
1008 end

```